生态研究前沿速览 _2026-03-18

Ecological & Environmental Sciences Journal Digest

2026 年 3 月 18 日（星期三）

共收录 300 篇最新研究 · 26 篇含中文深度解读 · 20 张图形摘要

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【综合顶刊】

1. Global patterns and drivers of soil microbial nitrogen and phosphorus use efficiency

【土壤微生物氮磷利用效率的全球格局与驱动因素】

□ Nature Communications | 2026-03-17 [Latest] | 重要性: 40 分

□ Abstract: Nature Communications, Published online: 17 March 2026; doi:10.1038/s41467-026-70602-0Soil microbes recycle nutrients, but their global nutrient use efficiency is poorly understood. This study finds microbial nitrogen-use efficiency is nearly twice phosphorus-use efficiency, driven by soil carbon content, and is lowest in tundra and boreal forests.

□ 中文摘要:

土壤微生物参与养分循环，但其全球养分利用效率尚不清楚。本研究发现微生物氮利用效率几乎是磷利用效率的两倍，受土壤碳含量驱动，揭示了全球土壤微生物生态化学计量学的普遍规律和驱动机制。

□ 深度解读:

本研究发表于 Nature Communications，首次在全球尺度上系统量化了土壤微生物的氮磷利用效率及其空间格局，填补了土壤生物地球化学循环研究的重要空白。研究揭示微生物氮利用效率普遍高于磷利用效率，且这一差异受土壤有机碳含量调控，体现了生态化学计量学的全球普遍规律。这一发

现对于构建和校验地球系统模型中的氮磷循环模块至关重要，也为预测气候变化下土壤养分动态提供了理论基础，对中国黄土高原、青藏高原等地区的施肥管理和碳氮耦合研究有直接指导意义。

☐ DOI: https://doi.org/10.1038/s41467-026-70602-0

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2. Quantifying the effects of response diversity dynamics on ecosystem stability

【量化响应多样性动态对生态系统稳定性的影响】

☐ Nature Communications | 2026-03-17 [Latest] | 重要性: 40 分

☐ Abstract: Nature Communications, Published online: 17 March 2026; doi:10.1038/s41467-026-70192-xSpecies' varied responses to environmental change help stabilize ecosystems, yet quantifying these dynamics remains challenging. This study reveals that plankton in Lake Geneva exhibit time-varying response diversity that stabilizes biomass within but not across trophic levels.

☐ 中文摘要:

物种对环境变化的不同响应有助于维持生态系统稳定性，但量化这些动态变化仍具挑战性。本研究揭示日内瓦湖浮游生物表现出随时间变化的响应多样性，这种多样性通过降低种群同步性来缓冲气候波动，增强生态系统稳定性。

☐ 深度解读:

本研究发表于 Nature Communications，以日内瓦湖 30 年浮游生物长期监测数据为基础，首次量化了响应多样性的时间动态性，揭示生态系统稳定性背后的功能机制。研究发现物种间对气候变化的差异化响应会随季节和年际变化动态波动，且这种动态变化本身对维持生态系统功能稳定至关重要。该发现为理解物种多样性与生态系统稳定性的关系提供了新的时间动态视角，对生物多样性保护的生态系统服务评估具有重要方法论意义。

☐ DOI: https://doi.org/10.1038/s41467-026-70192-x

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3. Marine conservation cities: a model for ocean governance

【海洋保护城市：海洋治理的新模式】

☐ Nature | 2026-03-17 [Latest] | 重要性: 38 分

☐ Abstract: Nature, Published online: 17 March 2026; doi:10.1038/d41586-026-00864-7Marine conservation cities: a model for ocean governance

☐ 中文摘要:

本文提出海洋保护城市概念，作为海洋治理的新模型，探讨城市在推动海洋保护中的主体作用，尤其是在国家层面海洋治理框架不足的情况下，城市如何通过本地政策和行动填补海洋保护的治理缺口。

☐ 深度解读:

本研究发表于 Nature，提出了一个创新性的海洋治理框架，将城市（尤其是沿海城市）定位为海洋保护的新行动主体。研究梳理了全球沿海城市在海洋保护中的现有实践，总结了城市尺度海洋保护的关键要素和成功案例。这一框架对中国沿海城市推进海洋保护建设、实现海洋强国战略具有重要参考价值，也与中国积极参与推动 30x30 海洋保护目标的国际承诺高度契合。

☐ DOI: https://doi.org/10.1038/d41586-026-00864-7

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4. Accelerated discovery of cell migration regulators using label-free deep learning – based automated tracking

☐ Science Advances | 2026-03-11T07:00:00Z [Latest] | 重要性: 37 分

☐ DOI: https://doi.org/10.1126/sciadv.aea1492

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5. Unravelling the Ru-promoted dynamic evolution of Cobalt hydroxide during nitrate reduction towards ammonia production

☐ Nature Communications | 2026-03-17 [Latest] | 重要性: 37 分

☐ Abstract: Nature Communications, Published online: 17 March 2026; doi:10.1038/s41467-026-70531-yThe unclear structural evolution mechanism of Cobalt-based materials restricts the long-term applications of nitrate reduction for ammonia synthesis. Here, the authors report a Ru-promoted dynamic evolution of cobalt hydroxides that can stably produce ammonia at high current densities.

☐ DOI: https://doi.org/10.1038/s41467-026-70531-y

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6. Nine changes needed to deliver a radical transformation in biodiversity measurement

【实现生物多样性测量根本性变革所需的九项改变】

☐ PNAS | 2026-03-04T08:00:00Z [Latest] | 重要性: 36 分

☐ Abstract: Biodiversity is declining in many parts of the world. Biological diversity measurement and monitoring are fundamental to the assessment of the causes and consequences of environmental changes, identification of key areas for the protection of biodiversity ...

☐ 中文摘要:

生物多样性在世界许多地方正在下降。生物多样性的测量和监测是评估环境变化原因和后果、识别生物多样性保护关键区域的基础。本文提出需要九项根本性改变，以实现生物多样性测量方法的革命性突破，支撑昆明-蒙特利尔全球生物多样性框架的实施。

☐ 深度解读:

本研究发表于 PNAS，针对当前生物多样性测量体系的系统性缺陷，提出九项变革性建议，涵盖从局地到全球尺度的测量方法、标准化框架、数据共享协议和新技术整合（eDNA、卫星遥感、公民科学）。研究的时机与 2030 年全球生物多样性保护目标高度契合，对中国推进昆明-蒙特利尔框架下 30x30 目标具有直接指导价值，也为中国生物多样性监测体系的现代化升级提供了理论框架。

☐ DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2519345123

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7. Stochasticity in mammalian cell growth rates drives cell-to-cell variability independently of cell size and divisions

☐ PNAS | 2026-03-03T08:00:00Z [Latest] | 重要性: 34 分

☐ Abstract: SignificanceGenetically identical cells can grow at different rates. This cell-to-cell growth variability can influence, for example, tumor biology, drug efficacy, and immune responses. Yet, the origin of this variability among genetically identical cells ...

☐ DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2516372123

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8. Cancer interception with KRAS inhibitors in preclinical models of pancreatic ductal adenocarcinoma

☐ Science | 2026-03-12T06:00:03Z [Latest] | 重要性: 32 分

☐ Abstract: Science, Volume 391, Issue 6790, Page 1161-1166, March 2026.

☐ DOI: https://doi.org/10.1126/science.aec7929

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9. Programmable genome editing in human cells using RNA-guided bridge recombinases

☐ Science | 2026-03-12T07:00:00Z [Latest] | 重要性: 32 分

☐ DOI: https://doi.org/10.1126/science.adz1884

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10. A persistent-range hydrogen-bonded gel polymer electrolyte enabling wide-temperature and recyclable lithium metal batteries

☐ Science Advances | 2026-03-11T07:00:00Z [Latest] | 重要性: 32 分

☐ DOI: https://doi.org/10.1126/sciadv.adz1014

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11. The Kananaskis Wildfire Charter: a good start

☐ Nature Communications | 2026-03-17 [Latest] | 重要性: 32 分

☐ Abstract: Nature Communications, Published online: 17 March 2026; doi:10.1038/s41467-026-70040-yThe Kananaskis Wildfire Charter marks an important step toward addressing increasingly extreme wildfires. Attending to complex drivers, supporting Indigenous and local fire stewardship, and shifting toward integrated fire management will help ensure equitable, effective, and ecologically grounded governance and management.

☐ DOI: https://doi.org/10.1038/s41467-026-70040-y

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12. Vertical substitution strategy to enable cooperation between spin–orbit coupling and transition dipoles for organic phosphorescence

☐ Nature Communications | 2026-03-17 [Latest] | 重要性: 32 分

☐ Abstract: Nature Communications, Published online: 17 March 2026; doi:10.1038/s41467-026-70371-wMultiple vertical hetero-substitutions on a conjugated chromophore enable strong cooperatively between the transition dipole moment and spin – orbit coupling in the higher-order excited states, which enhances the organic phosphorescence process.

☐ DOI: https://doi.org/10.1038/s41467-026-70371-w

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13. Profile of David Baker, Demis Hassabis, and John Jumper: 2024 Nobel laureates in chemistry

☐ PNAS | 2026-02-27T08:00:00Z [Latest] | 重要性: 32 分

☐ Abstract: The 2024 Nobel Prize in Chemistry recognized revolutionary computational success in predicting protein structures from amino acid sequences and for designing amino acid sequences that fold into desired protein structures. The ultimate accomplishment ...

☐ DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2422539123

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14. Persistent adaptation through dual-timescale regulation of ion channel properties

☐ PNAS | 2026-03-02T08:00:00Z [Latest] | 重要性: 32 分

☐ Abstract: SignificanceSome neurons exhibit a remarkable property: they recover more quickly from repeated challenges even though their baseline activity appears unchanged. How such history-dependent improvement in recovery arises is unclear. Using a computational ...

☐ DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2530340123

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15. Evaluating SSR marker transferability and plastid barcode variation in native Populus and Salix species of Türkiye

☐ PeerJ | 2026-03-17 [Latest] | 重要性: 23 分

☐ Abstract: Populus and Salix species are key components of forest and riparian ecosystems and hold substantial economic value in forestry, bioenergy, and restoration practices. In Türkiye, these genera are widely distributed across diverse ecological zones, yet regional molecular data remain limited. In this study, I examined genetic relationships among six native species—Populus nigra L., Populus alba L., Populus tremula L., Populus euphratica Olivier, Salix alba L., and Salix caprea L.—using an integrative molecular approach. Twelve nuclear microsatellite (simple sequence repeats; SSR) markers and three plastid DNA barcode regions (matK, rbcL, trnH−psbA) were employed to assess marker performance, genetic variation, and species-level relationships. The analyses revealed clear genetic differentiatio

☐ DOI: https://peerj.com/articles/20936

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16. Environmental drivers of Catostylus tagi polyp survival and reproduction: unlocking the role of temperature and salinity, supported with citizen science data

☐ PeerJ | 2026-03-17 [Latest] | 重要性: 19 分

☐ Abstract: Background Catostylus tagi is a scyphozoan jellyfish native to Portuguese waters. While its life cycle is known, the environmental conditions that support polyp survival, trigger strobilation, and promote asexual reproduction remain unclear. Field observations from the citizen science GelAvista project indicate that C. tagi occurs year-round in the Tagus estuary, suggesting tolerance to broad temperature and salinity ranges. However, polyps and ephyrae have not been observed in the wild, and their natural habitats and environmental preferences remain unknown. This study aims to fill these knowledge gaps by investigating how temperature and salinity affect C. tagi polyps under controlled conditions and integrating these results with field data. Methods Ninety-six polyps were cultured for 71

☐ DOI: https://peerj.com/articles/20862

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17. The impact of stability considerations on genetic fine-mapping

☐ eLife | Mon, 16 Mar 2026 00: [Latest] | 重要性: 17 分

☐ Abstract: Fine-mapping methods, which aim to identify genetic variants responsible for complex traits following genetic association studies, typically assume that sufficient adjustments for confounding within the association study cohort have been made, for example, through regressing out the top principal components (i.e., residualization). Despite its widespread use, however, residualization may not completely remove all sources of confounding. Here, we propose a complementary stability-guided approach that does not rely on residualization, which identifies consistently fine-mapped variants across different genetic backgrounds or environments. Simulations show that stability guidance neither outperforms nor underperforms residualization, but each approach picks up different variants considerably o

☐ DOI: https://elifesciences.org/articles/88039

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18. Air and soil warming have different effects on soil organic carbon storage

☐ Communications Earth & Environment | 2026-03-16 [Latest] | 重要性: 14 分

☐ Abstract: Communications Earth & Environment, Published online: 16 March 2026; doi:10.1038/s43247-026-03367-5Soil organic carbon storage decreases with soil warming due to temperature-driven increases in decomposition rate, whereas its responses to air warming can vary depending on net primary production and soil moisture, according to a mechanistic model and 327 data points across 139 sites worldwide.

☐ DOI: https://doi.org/10.1038/s43247-026-03367-5

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19. Reduced carbon outflow from a Floridian mangrove estuary up to two years after a hurricane

☐ Communications Earth & Environment | 2026-03-17 [Latest] | 重要性: 12 分

☐ Abstract: Communications Earth & Environment, Published online: 17 March 2026; doi:10.1038/s43247-026-03249-wMangroves in Florida show a substantial and sustained decrease in dissolved organic and inorganic carbon outwelling following the passage of a category 3-4 hurricane, according to analysis of a high-resolution carbon outwelling timeseries from Everglades National Park.

☐ DOI: https://doi.org/10.1038/s43247-026-03249-w

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20. Subspecies-specific haplotype signatures for customizing blanchability in groundnut (Arachis hypogaea L.) via haplotype-based breeding

☐ Communications Biology | 2026-03-15 [Latest] | 重要性: 9 分

☐ Abstract: Communications Biology, Published online: 15 March 2026; doi:10.1038/s42003-026-09850-1Genomic investigation of the ICRISAT mini-core collection revealed subspecies-specific favorable haplotypes and diagnostic markers associated with blanchability, the ease of seed coat removal, a key processing trait in groundnut.

☐ DOI: https://doi.org/10.1038/s42003-026-09850-1

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21. Dietary sulfur amino acid restriction elicits a cold-like transcriptional response in inguinal but not epididymal white adipose tissue of male mice

☐ eLife | Tue, 17 Mar 2026 00: [Latest] | 重要性: 9 分

☐ Abstract: About 1 billion people are living with obesity worldwide. GLP-1-based drugs have massively transformed care, but long-term consequences are unclear in part due to reductions in energy expenditure with ongoing use. Diet-induced thermogenesis (DIT) and cold exposure (CE) raise EE via brown adipose tissue (BAT) activation and beiging of white adipose tissue (WAT). Methionine restriction (MetR) is a candidate DIT stimulus, but its EE effect has not been benchmarked against CE, nor have their tissue-level interactions been defined. In a 2×2 design (Control vs. MetR; room temperature, RT: 22°C vs. CE: 4°C for 24 hr), we used male C57BL/6 N mice to benchmark MetR-induced thermogenesis against CE and mapped how diet and temperature interact across tissues. Bulk RNA-seq profiled liver, iBAT, iWAT,

☐ DOI: https://elifesciences.org/articles/108825

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22. Cardenolide toxin diversity impacts monarch butterfly growth and sequestration

☐ eLife | Mon, 16 Mar 2026 00: [Latest] | 重要性: 8 分

☐ Abstract: In coevolutionary interactions, host plants accrue novel chemical defenses that specialist herbivores counter by detoxification and sometimes sequestration. We recently found unusual nitrogen- and sulfur-containing (N,S-) cardenolides in some milkweeds—highly toxic compounds that monarch butterflies (Danaus plexippus) detoxify during sequestration. We hypothesized that the N,S-cardenolides in Asclepias curassavica (uscharin and voruscharin) would reduce caterpillar performance and sequestration more than other abundant related cardenolides (15-hydroxy-calotropin, frugoside, calactin). Cardenolides generally increased feeding relative to controls, but voruscharin was not stimulatory and substantially reduced growth efficiency. Exposure to either N,S-cardenolide produced the lowest sequestra

☐ DOI: https://elifesciences.org/articles/109003

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23. Oceanic forcing of patagonian ice sheet variability over the last eight glacial cycles

☐ Communications Earth & Environment | 2026-03-17 [Latest] | 重要性: 8 分

☐ Abstract: Communications Earth & Environment, Published online: 17 March 2026; doi:10.1038/s43247-026-03387-1Sea surface temperatures in the Southeast Pacific controlled the Patagonian Ice Sheet's mass balance between glacials and interglacials over the past ~800 thousand years, according to analysis of a marine sediment record from the southern Chilean margin.

☐ DOI: https://doi.org/10.1038/s43247-026-03387-1

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24. Pathways for including non-carbon dioxide aviation climate effects in the European Emission Trading System

☐ Communications Earth & Environment | 2026-03-17 [Latest] | 重要性: 8 分

☐ Abstract: Communications Earth & Environment, Published online: 17 March 2026; doi:10.1038/s43247-026-03265-wFor the inclusion of aviation non-carbon dioxide climate effects in the European Union Emission Trading System, uncertainties in atmospheric processes can be addressed through careful calculation and risk assessment.

☐ DOI: https://doi.org/10.1038/s43247-026-03265-w

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25. Extent of alveolar collapse in expiratory CT as a prognostic marker in idiopathic pulmonary fibrosis

☐ PLOS ONE | 2026-03-17T14:00:00Z [Latest] | 重要性: 6 分

☐ Abstract: by Sarah C. Scharm, Cornelia Schaefer-Prokop, Anton Schreuder, Jonathan Ehmig, Anna Hunkemöller, Jan Fuge, Benjamin Seeliger, Jonas Schupp, Frank K. Wacker, Hoen-oh Shin To evaluate whether distribution measures of CT-based attenuation histograms in inspiration and expiration can indicate alveolar collapse and serve as a predictive marker in patients with idiopathic pulmonary fibrosis (IPF). This single-center retrospective longitudinal study analyzed CT scans of IPF patients in inspiration and expiration. The patient population was divided into two subgroups based on their status 3 years after baseline CT (death or transplantation versus clinical surveillance). Attenuation histograms in inspiration and expiration were created and analyzed. A Mann-Whitney U test was conducted to assess the

☐ DOI: https://journals.plos.org/plosone/article

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26. Controls of root-system overlap on hillslope stability

☐ Communications Earth & Environment | 2026-03-16 [Latest] | 重要性: 6 分

☐ Abstract: Communications Earth & Environment, Published online: 16 March 2026; doi:10.1038/s43247-025-03012-7Vegetation density influences slope stability with higher root-system overlaps delaying landslide initiation and smaller landslide areas at moderate densities, according to lab flume experiments using pea plants with different densities.

☐ DOI: https://doi.org/10.1038/s43247-025-03012-7

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27. The secretory protein, CLCF1, improves cholestatic liver disease by inhibiting hepatic bile acid synthesis and promoting bile acid excretion

☐ Communications Biology | 2026-03-16 [Latest] | 重要性: 4 分

☐ Abstract: Communications Biology, Published online: 16 March 2026; doi:10.1038/s42003-026-09847-wUsing chemically and genetically induced cholestatic liver disease in male mice, this study reveals that CLCF1 alleviates disease features by suppressing bile acid synthesis in the liver and promoting its excretion via gut microbiota remodelling.

☐ DOI: https://doi.org/10.1038/s42003-026-09847-w

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28. Where design meets biology: an interview on SPIKA, a prototype for microbe-mediated architecture

☐ Communications Biology | 2026-03-14 [Latest] | 重要性: 4 分

☐ Abstract: Communications Biology, Published online: 14 March 2026; doi:10.1038/s42003-026-09795-5This Q&A explores the interdisciplinary advances in the EU Pathfinder project Microbial Hydroponics (Mi-Hy) and its SPIKA installation - a functional prototype installed at the Milan Triennale 2025. In this prototype, microbial electrochemistry, engineered biomineralization, and metabolic programming converge. Researchers Ezgi Öğün Ramalhete, Işıl Yücel, and Jorge Barriuso discuss how microbial consortia can reshape the foundations of the built environment by introducing a platform for sustainable agriculture. This might enable a transition from resource-consuming architecture to circular, living infrastructure.

☐ DOI: https://doi.org/10.1038/s42003-026-09795-5

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29. Effect of fatigue on neuromuscular adaptations in endurance-trained and recreationally active males

☐ PeerJ | 2026-03-17 [Latest] | 重要性: 4 分

☐ Abstract: Background Neuromuscular fatigue can be characterized by an exercise-induced reduction in force-generating capacity involving both neural and muscular mechanisms. Previous research has suggested that the functional organization of the neuromuscular system differs between endurance-trained individuals and sedentary or recreationally active individuals. This difference may lead to distinct neuromuscular responses to fatigue. The aim of this study was to compare neuromuscular fatigue responses between endurance-trained (ET) and recreationally active (RA) males during a sustained submaximal isometric knee extension contraction. Methods Eleven ET and 11 RA participants performed maximal voluntary isometric contractions (MVIC) of the knee extensors (KE), followed by a trapezoidal contraction (as

☐ DOI: https://peerj.com/articles/20979

【生态学核心】

30. Responses of African Savanna Trees to Large Herbivore Extinction and Rewilding

【非洲热带稀树草原树木对大型食草动物灭绝与再野化的响应】

☐ Ecology Letters | Thu, 12 Mar 2026 11: [Latest] | 重要性: 53 分

☐ 中文摘要:

本研究检验了非洲热带稀树草原树木对大型食草动物灭绝（历史丧失）和再野化（重新引入）的生态响应，揭示顶级消费者对草原生态系统结构和功能的级联调控效应及其可逆性。

☐ 深度解读:

本研究发表于 Ecology Letters，通过比较有无大型食草动物（大象、犀牛等）区域的树木种群结构，量化了顶级食草动物对热带稀树草原树木群落的调控效应及其在再野化后的恢复动态。研究发现食草动物通过改变植被结构根本性地塑造草原景观，且这种影响在再野化后表现出一定的可逆性但存在时滞。结果对全球再野化项目的设计和预期效果评估具有重要启示，也为非洲国家公园大型动物恢复项目的科学决策提供依据。

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70360

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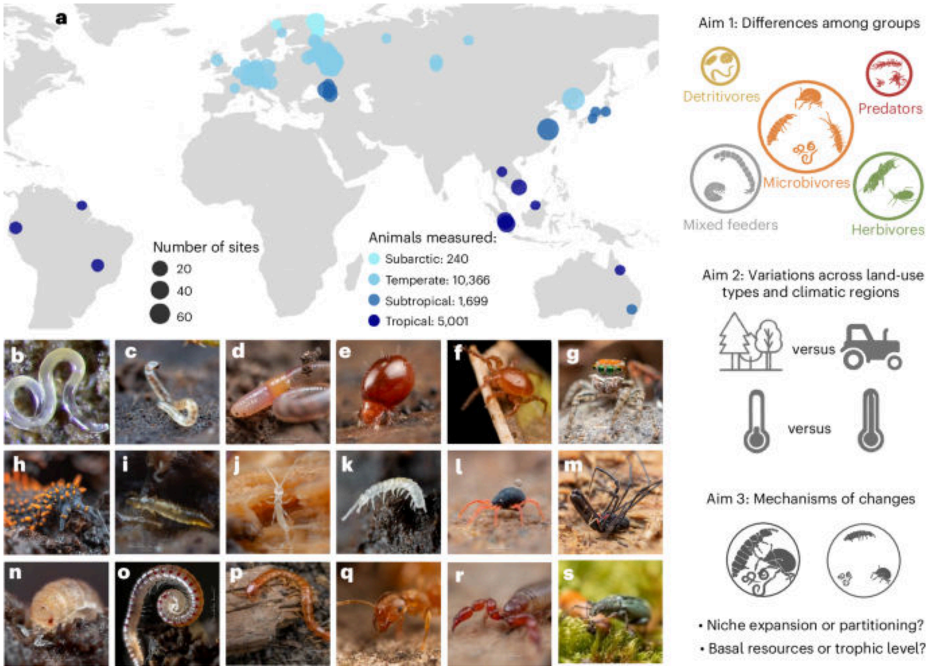
31. Greater trophic diversity of soil animal communities under agricultural land use and tropical climate

【农业用地和热带气候下土壤动物群落具有更高的营养多样性】

☐ Nature Ecology & Evolution | 2026-03-16 [Latest] | 重要性: 46 分

☐ Graphical Abstract:



☐ Abstract: Nature Ecology & Evolution, Published online: 16 March 2026; doi:10.1038/s41559-026-03014-4Soil fauna is an important but often neglected component of terrestrial food webs. Here the authors use a large dataset of stable isotope observations to analyse how soil animal trophic diversity varies across climates and land-use types and identify potential biotic mechanisms.

☐ 中文摘要:

土壤动物是陆地食物网的重要但常被忽视的组分。本研究利用大规模稳定同位素观测数据集，分析土壤动物营养多样性如何随土地利用、气候和土壤性质的变化而变化，揭示农业土地利用和热带气候对土壤动物群落结构的塑造作用。

☐ 深度解读:

本研究发表于 Nature Ecology & Evolution，利用全球最大规模的土壤动物稳定同位素数据集，从营养生态学角度揭示了农业干扰和热带气候对土壤生物多样性的影响规律。研究发现农业土地利用虽然通常被认为会降低土壤生物多样性，但实际上可能增加土壤动物的营养功能多样性，这对理解农业生态系统中物质循环效率具有重要意义。该发现挑战了农业简化食物网的固有认知，为评估农业可持续性提供了新的功能多样性指标体系。

☐ DOI: https://doi.org/10.1038/s41559-026-03014-4

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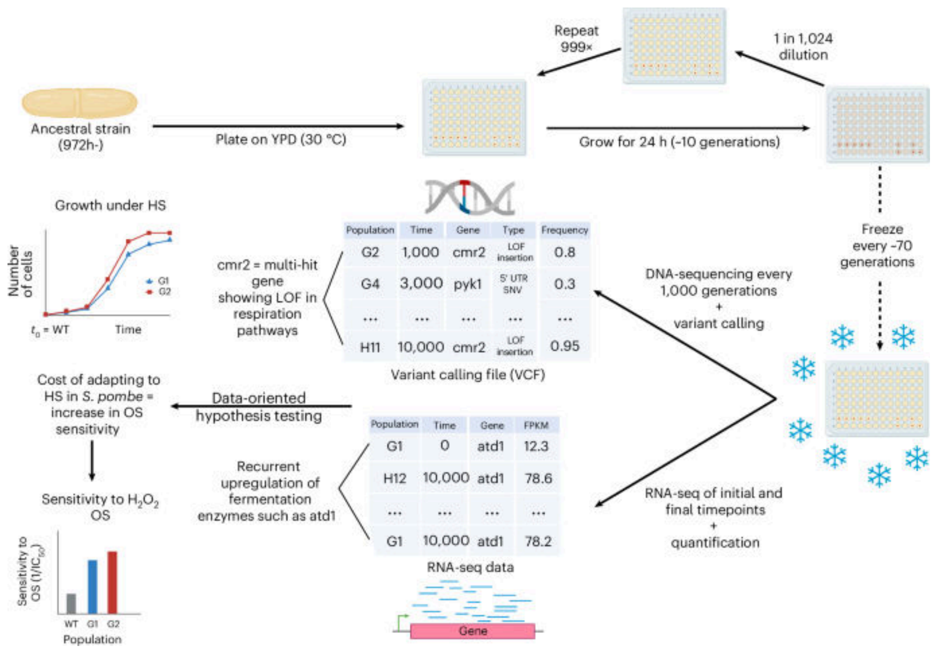
32. Parallel but distinct adaptive routes in the budding and fission yeasts after 10,000 generations of experimental evolution

☐ Nature Ecology & Evolution | 2026-03-13 [Latest] | 重要性: 40 分

☐ Graphical Abstract:



☐ Abstract: Nature Ecology & Evolution, Published online: 13 March 2026; doi:10.1038/s41559-026-03017-1Experimental evolution of fission yeast (Schizosaccharomyces pombe) in the same environment as a previous experiment with budding yeast (Saccharomyces cerevisiae) reveals parallel evolution but distinct molecular mechanisms and targets of adaptation in the two species.

☐ DOI: https://doi.org/10.1038/s41559-026-03017-1

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33. Biomass and Functional Traits of Plants and Soil Carbon-to-Nitrogen Ratio Jointly Control the Effect of Living Roots on Soil Carbon Decomposition

【植物生物量与功能性状及土壤碳氮比共同调控活根对土壤碳分解的影响】

☐ Global Change Biology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 40 分

☐ 中文摘要:

本研究揭示植物生物量与功能性状（如根系特征）以及土壤碳氮比如何共同调控活根对土壤有机碳分解的激发效应，为理解植物-土壤碳循环相互作用提供了新的机制认识。

☐ 深度解读:

本研究发表于 Global Change Biology，通过控制实验和大样本分析，阐明了根系激发效应的关键调控因素，揭示植物功能性状（特别是根系碳氮比和根际沉积物化学组成）与土壤碳氮比的交互作用如何决定活根对土壤碳分解的促进或抑制效应。这一发现对于准确模拟植物-土壤碳循环耦合过程至关重要，为改进碳循环模型和预测不同植被类型（如人工林 vs 天然林）下土壤碳库动态提供了重要参数化依据。

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70803

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34. Structural characteristics mediate forest mitigation potential against climate change and biodiversity loss

【森林结构特征介导其对气候变化和生物多样性丧失的缓解潜力】

☐ Ecological Applications | Mon, 16 Mar 2026 17: [Latest] | 重要性: 38 分

☐ 中文摘要:

本研究探讨森林结构特征如何调控森林对气候变化和生物多样性丧失的缓解潜力，揭示林分结构对森林碳汇功能和生物多样性维持的关键调控作用。

☐ 深度解读:

本研究发表于 Ecological Applications，从森林结构功能角度揭示了结构复杂性对森林气候和生物多样性双重缓解功能的调控机制。研究发现多层次、多年龄组的结构复杂型森林不仅碳储量更高，也能支撑更丰富的生物多样性，体现了森林结构保护的协同效益。这一发现对中国天然林保护工程和人工林近自然化改造具有直接指导意义，支持推进碳汇林建设从单一树种高密度人工林转向近自然混交林的政策方向。

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70211

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35. Genetic ancestry affects cancer immunity

☐ Nature Ecology & Evolution | 2026-03-11 [Latest] | 重要性: 36 分

☐ Abstract: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03012-6Genetic ancestry affects cancer immunity

☐ DOI: https://doi.org/10.1038/s41559-026-03012-6

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36. Much Larger Whole-Profile Soil Organic Carbon Stocks on the Qinghai-Tibet Plateau Than Previously Reported

☐ Global Change Biology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 36 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70808

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37. Widespread Higher Soil Respiration Rates at Nighttime Than Daytime Across Global Forest Ecosystems

☐ Global Change Biology | Fri, 13 Mar 2026 23: [Latest] | 重要性: 36 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70798

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38. Top-Down and Bottom-Up Processes Jointly Explain Mesopredator Movement and Foraging Ecology

☐ Ecology Letters | Sat, 14 Mar 2026 11: [Latest] | 重要性: 34 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70364

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39. The Leading Edge Matters Too: Fitness and the Expression of Adaptive Differentiation Are Greatest at the High-Elevation Edge of a Species' Range

☐ Ecology Letters | Thu, 12 Mar 2026 11: [Latest] | 重要性: 34 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70329

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40. Biodiversity Trends Show an Excess of Both Near Stasis and of Very Large Change

【生物多样性趋势显示近乎停滞和极大变化同时过剩】

☐ Ecology Letters | Wed, 11 Mar 2026 17: [Latest] | 重要性: 34 分

☐ 中文摘要:

本研究分析全球生物多样性时间序列数据，发现生物多样性变化趋势存在两端极值过剩的特征：既有大量地点表现出近乎静止的状态，也有大量地点经历了极端的变化，这种双峰分布对理解全球生物多样性危机的复杂性具有重要意义。

☐ 深度解读:

本研究发表于 Ecology Letters，通过分析全球生物多样性数据库，揭示生物多样性时间变化趋势并非呈现简单的正态分布，而是存在双峰型分布，即同时出现大量近乎不变和剧烈变化的生态系统。这一发现挑战了生物多样性普遍下降的简单化叙事，表明不同生态系统对人类活动和气候变化的响应存在高度异质性。对生物多样性监测指标设计和全球生物多样性评估框架（IPBES）的改进具有重要方法论启示。

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70353

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41. Plasticity in prostate cancer

☐ Nature Ecology & Evolution | 2026-03-11 [Latest] | 重要性: 34 分

☐ Abstract: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03024-2Plasticity in prostate cancer

☐ DOI: https://doi.org/10.1038/s41559-026-03024-2

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42. Drift shapes norovirus diversity

☐ Nature Ecology & Evolution | 2026-03-11 [Latest] | 重要性: 34 分

☐ Abstract: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03022-4Drift shapes norovirus diversity

☐ DOI: https://doi.org/10.1038/s41559-026-03022-4

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43. Deep Coring Shows That Mangrove Sediments in Matang (Malaysia) Store up to Five Times More Carbon Than Previously Estimated

☐ Global Change Biology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 34 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70773

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44. Anaemic Streams: Iron and Essential Trace Metals Frequently Limit Primary Producer Biomass

☐ Ecology Letters | Sun, 08 Mar 2026 10: [Latest] | 重要性: 32 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70357

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45. Large Potential for CH4 Mitigation and Yield Improvement in China's Paddies Through Locally Optimized N Management

☐ Global Change Biology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 30 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70801

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46. Assessing the environmental and dispersal-related drivers of an invasive aquatic plant in Great Lakes coastal wetlands

☐ Ecological Applications | Thu, 12 Mar 2026 00: [Latest] | 重要性: 14 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70206

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47. Total herbivore exclusion alters plant species dominance and trait expression in endangered grassy woodlands

☐ Ecology | Mon, 16 Mar 2026 20: [Latest] | 重要性: 10 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70350

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48. Effects of temperature and browning on the functional response of a freshwater top predator

☐ Journal of Animal Ecology | Mon, 16 Mar 2026 22: [Latest] | 重要性: 10 分

☐ Abstract: Journal of Animal Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70233

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49. Climate-linked evolution and genetics in a warming Arctic

☐ Ecological Monographs | Mon, 16 Mar 2026 05: [Latest] | 重要性: 10 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70053

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50. Environmental filtering drives mycorrhizal tree dominance across a soil fertility gradient in a temperate forest

☐ Journal of Ecology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 8 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70298

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51. Enhanced below-ground functioning is associated with higher plant resistance against drought: Implications for ecosystem functions

☐ Journal of Ecology | Fri, 13 Mar 2026 00: [Latest] | 重要性: 8 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70289

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52. Temperature controls bryophyte-associated nitrogen fixation in super-humid temperate forests in New Zealand

☐ Ecology | Mon, 16 Mar 2026 20: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70343

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53. Mammal defaunation leads to biotic homogenization of plant communities in tropical rain-forests

☐ Ecology | Mon, 16 Mar 2026 20: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70341

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54. Eco-evolutionary dynamics of partially migratory metapopulations in spatially and seasonally varying environments

☐ Journal of Animal Ecology | Thu, 12 Mar 2026 22: [Latest] | 重要性: 8 分

☐ Abstract: Journal of Animal Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70240

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55. Extending plant water-use strategies to flowers: Evidence from trait correlations across plant organs

☐ Functional Ecology | Tue, 10 Mar 2026 21: [Latest] | 重要性: 8 分

☐ Abstract: Functional Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70293

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56. The post-fire recovery of soil seed banks along a fire severity gradient in an Australian threatened mesic forest

☐ Functional Ecology | Fri, 06 Mar 2026 03: [Latest] | 重要性: 8 分

☐ Abstract: Functional Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70296

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57. Year-round rhythms: Alpine plant species modulate soil and microbial dynamics during the growing season and under the snow

☐ Ecological Monographs | Wed, 11 Mar 2026 22: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70055

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58. A transcontinental experiment elucidates (mal)adaptation of a cosmopolitan plant to climate in space and time

☐ Ecological Monographs | Sun, 15 Feb 2026 16: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70043

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59. Does restoring apex predators to food webs restore ecosystems? Reply

☐ Ecological Monographs | Tue, 03 Feb 2026 23: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70049

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60. Modeling species co-occurrence effects to inform invasive barred owl management and recovery of the northern spotted owl

☐ Ecological Applications | Thu, 12 Mar 2026 00: [Latest] | 重要性: 8 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70195

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61. Applying the maximum entropy principle to neural networks enhances multi-species distribution models

☐ Methods in Ecology and Evolution | Mon, 16 Mar 2026 09: [Latest] | 重要性: 8 分

☐ Abstract: Methods in Ecology and Evolution, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70262

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62. Automated counting of fish in moving diver-operated videos (DOV) for biodiversity assessments

☐ Methods in Ecology and Evolution | Thu, 12 Mar 2026 23: [Latest] | 重要性: 8 分

☐ Abstract: Methods in Ecology and Evolution, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70283

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63. Century-long trends in plant diversity of temperate mountain vegetation are modulated along elevation gradient

☐ Journal of Ecology | Thu, 12 Mar 2026 04: [Latest] | 重要性: 6 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70292

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64. Drivers and conservation impacts of innovative tree nesting in an isolated island population of the red-billed chough

☐ Ecology | Mon, 16 Mar 2026 20: [Latest] | 重要性: 6 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70344

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65. On traits matching and the modular organization of food web and occurrence networks

☐ Journal of Animal Ecology | Mon, 16 Mar 2026 22: [Latest] | 重要性: 6 分

☐ Abstract: Journal of Animal Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70234

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66. Functional and phylogenetic clustering shape rodent co-occurrence via multi-scale mechanisms in a subtropical forest

☐ Journal of Animal Ecology | Fri, 06 Mar 2026 01: [Latest] | 重要性: 6 分

☐ Abstract: Journal of Animal Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70237

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67. Nitrogen deposition does not exacerbate phosphorus limitation of rhizosphere microbes in subalpine forests

☐ Functional Ecology | Mon, 09 Mar 2026 01: [Latest] | 重要性: 6 分

☐ Abstract: Functional Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70297

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68. An invasive shrub drives carbon uptake in New Zealand montane tussock grasslands, with strongest effects early in the growing season

☐ Functional Ecology | Thu, 05 Mar 2026 19: [Latest] | 重要性: 6 分

☐ Abstract: Functional Ecology, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70291

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69. Biological soil crusts enhance the role of soil seed banks in semi-arid ecosystems

☐ Journal of Applied Ecology | Mon, 16 Mar 2026 02: [Latest] | 重要性: 6 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70327

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70. Restoration affects the temporal stability of biodiversity and productivity in degraded grass-lands

☐ Journal of Applied Ecology | Mon, 16 Mar 2026 00: [Latest] | 重要性: 6 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70299

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71. How to eradicate an invasive forest pest without clear-cutting

☐ Journal of Applied Ecology | Wed, 11 Mar 2026 22: [Latest] | 重要性: 6 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70318

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72. Inferring breeding phenology and reproductive success from the emergence of juveniles in population monitoring

☐ Oikos | Sun, 15 Mar 2026 00: [Latest] | 重要性: 6 分

☐ Abstract: Oikos, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.11520

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73. Marine heatwave and keystone predator loss drive broad-scale decline and hinder recovery of a rocky intertidal kelp

☐ Ecological Applications | Mon, 16 Mar 2026 19: [Latest] | 重要性: 6 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70215

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74. Habitat complexity and prey composition shape an apex predator's habitat use across contrasting landscapes

☐ Ecography | Sun, 15 Mar 2026 23: [Latest] | 重要性: 6 分

☐ Abstract: Ecography, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08132

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75. ʻukbioprepr': an R package to support reproducible preparation of environmental data for biodiversity modelling in the UK

☐ Ecography | Thu, 12 Mar 2026 05: [Latest] | 重要性: 6 分

☐ Abstract: Ecography, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08413

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76. EcoViz: a tool for visual analysis and photorealistic rendering of forest landscape model simulations

☐ Ecography | Wed, 11 Mar 2026 00: [Latest] | 重要性: 6 分

☐ Abstract: Ecography, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08198

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77. Acoustic Indices Reveal Fundamental Differences in Daily Phenology of Tropical and Temperate Forest Soundscapes

☐ Global Ecology and Biogeography | Thu, 12 Mar 2026 21: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/geb.70224

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78. Ecoacoustics for context-rich direct and indirect trophic interaction data and ecological network construction

☐ Methods in Ecology and Evolution | Thu, 12 Mar 2026 23: [Latest] | 重要性: 6 分

☐ Abstract: Methods in Ecology and Evolution, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70288

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79. A novel approach for distinguishing noise and information in forest MLS-point clouds

☐ Methods in Ecology and Evolution | Thu, 12 Mar 2026 23: [Latest] | 重要性: 6 分

☐ Abstract: Methods in Ecology and Evolution, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70280

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80. CheckEM: An open-source toolkit for standardising, cleaning and visualising stereo-video fish survey data

☐ Methods in Ecology and Evolution | Thu, 12 Mar 2026 23: [Latest] | 重要性: 6 分

☐ Abstract: Methods in Ecology and Evolution, EarlyView.

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70287

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81. The role of the bole: Constraints on the remobilization of stem non-structural carbohydrates under experimental carbon limitation

☐ Journal of Ecology | Thu, 12 Mar 2026 03: [Latest] | 重要性: 4 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70286

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82. Initial tree census for the Paint Rock Forest Dynamics Plot, Alabama, USA

☐ Ecology | Sun, 15 Mar 2026 19: [Latest] | 重要性: 4 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70348

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83. Think globally, breed locally: Limited genetic impact of management in solitary bees (Osmia bicornis and Osmia cornuta)

☐ Journal of Applied Ecology | Wed, 11 Mar 2026 02: [Latest] | 重要性: 4 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70321

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84. Long-term monitoring reveals how pondscape connectivity shapes the early spread of a biological invasion

☐ Journal of Applied Ecology | Wed, 11 Mar 2026 02: [Latest] | 重要性: 4 分

☐ DOI: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70325

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85. Metabolic growth mechanisms and theoretical growth potential of global woody plant communities

☐ Ecological Monographs | Thu, 05 Feb 2026 16: [Latest] | 重要性: 4 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70048

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86. Cotton facilitates long-distance seed dispersal by functioning as nest material for birds

☐ Oikos | Sun, 15 Mar 2026 23: [Latest] | 重要性: 4 分

☐ Abstract: Oikos, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.12185

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87. Urban-driven homogenization of aquatic subsidy size structure cascades to riparian predator communities

☐ Oikos | Sun, 15 Mar 2026 23: [Latest] | 重要性: 4 分

☐ Abstract: Oikos, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.12080

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88. Seasonal variation of leaf functional traits in sub-Arctic plants

☐ Oikos | Sun, 15 Mar 2026 23: [Latest] | 重要性: 4 分

☐ Abstract: Oikos, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.11849

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89. Consumer resilience suppresses the recovery of overgrazed ecosystems

☐ Ecological Applications | Wed, 11 Mar 2026 00: [Latest] | 重要性: 4 分

☐ DOI: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70196

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90. Tracing the origins and evolution of nymphalid butterflies (Lepidoptera) in the Atlantic Forest

☐ Ecography | Sun, 15 Mar 2026 23: [Latest] | 重要性: 4 分

☐ Abstract: Ecography, EarlyView.

☐ DOI: https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08419

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91. Spatial Autocorrelation of Species Diversity and Distributions in Time and Across Spatial Scales

☐ Global Ecology and Biogeography | Mon, 16 Mar 2026 20: [Latest] | 重要性: 4 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/geb.70221

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92. Ecological and Life-History Correlates of Incubation Attentiveness Differ Between Female-Only and Biparentally Incubating Passerine Birds

☐ Global Ecology and Biogeography | Sun, 15 Mar 2026 21: [Latest] | 重要性: 4 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/geb.70225

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93. Forest Reorganisation After Natural Disturbance: A Synthesis

☐ Global Ecology and Biogeography | Fri, 13 Mar 2026 05: [Latest] | 重要性: 4 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/geb.70220

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94. Diversification and Evolutionary Dynamics in Tropical Montane Regions

☐ Global Ecology and Biogeography | Wed, 11 Mar 2026 20: [Latest] | 重要性: 4 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/geb.70218

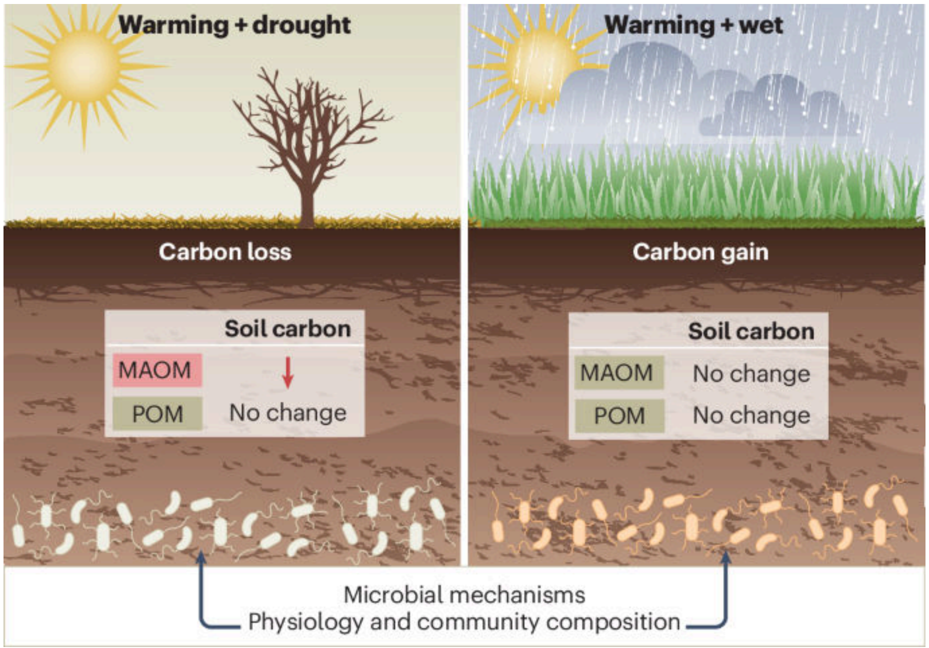
【环境与气候】

95. Dry soils lose more carbon when warm

【干燥土壤在气候变暖条件下碳损失更多】

☐ Nature Climate Change | 2026-03-13 [Latest] | 重要性: 57 分

☐ Graphical Abstract:



☐ Abstract: Nature Climate Change, Published online: 13 March 2026; doi:10.1038/s41558-026-02585-1The massive carbon store in soils is vulnerable to anthropogenic warming. Now, a study shows that climate-driven changes in precipitation can mediate soil carbon responses to warming, with drought amplifying soil carbon losses.

☐ 中文摘要:

土壤中储存着大量碳，这是陆地生态系统中最大的活性碳库。本文揭示气候变暖导致的降水变化会调控土壤碳对增温的响应，其中干旱会放大土壤碳损失。研究表明，在全球变暖背景下，气候引发的土壤湿度下降将显著加速土壤有机碳的分解，形成气候-碳循环正反馈。

☐ 深度解读:

本研究发表于 Nature Climate Change，揭示了气候变化影响土壤碳库的关键机制。研究发现降水减少诱发的土壤干旱会显著放大增温对土壤碳的损失效应，打破了以往认为干旱会抑制微生物活性从而保护碳的传统认知。这一发现对全球碳循环模型具有重要修正意义，提示现有气候模型可能低估了气候变暖的正反馈效应。对中国西北干旱半干旱区土壤碳库管理和碳中和目标实现具有直接政策指导价值。

☐ DOI: https://doi.org/10.1038/s41558-026-02585-1

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96. Sleep loss in a warming world

【变暖世界中的睡眠丧失】

☐ Nature Sustainability | 2026-03-13 [Latest] | 重要性: 57 分

☐ Graphical Abstract:

☐ Abstract: Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01778-yHigh temperatures disrupt sleep worldwide, with disproportionate impacts on older adults, women and populations in lower-income countries. A study uses climate change simulations to project future global sleep erosion and, in turn, the decline in childhood general cognitive ability and associated socioeconomic costs.

☐ 中文摘要:

高温在全球范围内扰乱睡眠，对老年人、女性及低收入国家人口的影响尤为严重。本研究利用气候变化模拟预测未来全球睡眠侵蚀的情景，揭示气候变化对人类健康的又一重要但常被忽视的影响路径。

☐ 深度解读:

本研究在 Nature Sustainability 上发表，将气候变化与人类睡眠健康相连接，开拓了气候健康影响研究的新维度。研究利用全球气候模拟数据量化了未来情景下睡眠时长和质量的损失，发现热带和亚热带地区（包括中国南部）面临最大风险。睡眠剥夺将引发认知损害、劳动生产率下降和慢性病发病率上升等级联效应。研究凸显了气候适应措施（城市绿化、建筑节能改造）对公共卫生的协同效益，为气候政策提供了新的健康代价量化依据。

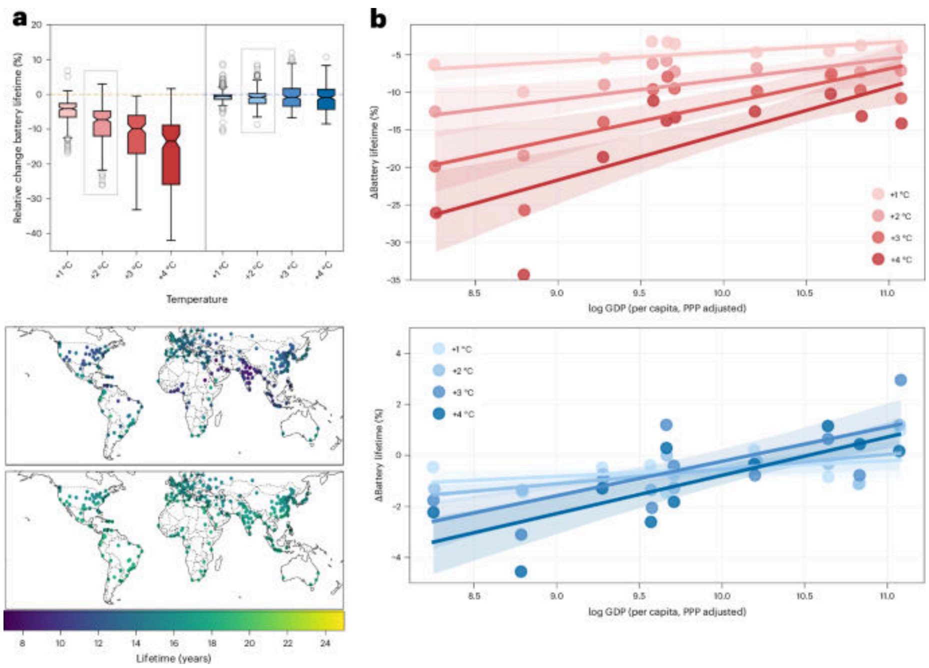
☐ DOI: https://doi.org/10.1038/s41893-026-01778-y

97. Technological advances mitigate the impact of climate change on electric vehicle battery lifetimes

【技术进步减缓气候变化对电动汽车电池寿命的影响】

☐ Nature Climate Change | 2026-03-16 [Latest] | 重要性: 53 分

☐ Graphical Abstract:



☐ Abstract: Nature Climate Change, Published online: 16 March 2026; doi:10.1038/s41558-026-02581-5We combined electric vehicle simulation and battery degradation models with high-resolution downscaled climate data for 300 global cities. Climate change was predicted to reduce battery lifetime by 8% on average for batteries manufactured between 2010 and 2018 versus 3% for batteries produced after 2019. Thus, technological advances in electric vehicle battery manufacturing demonstrate important climate adaptation co-benefits.

☐ 中文摘要:

研究将电动汽车仿真和电池退化模型与 300 个全球城市的高分辨率降尺度气候数据相结合，预测气候变化将平均缩短电池寿命 8%，但技术进步将在主要地区部分或完全抵消这一影响。

☐ 深度解读:

本研究发表于 Nature Climate Change，是首个系统量化气候变化对全球电动汽车电池寿命影响及技术对冲潜力的研究。研究利用多城市、多气候情景的大规模模拟，揭示炎热气候地区将是电池寿命受气候影响最严重的地区。这一发现对中国新能源汽车产业政策具有直接参考价值：既为研发面向热带亚热带气候的电池热管理技术提供了市场导向，也为电动汽车全生命周期碳核算提供了新的气候修正参数。

☐ DOI: https://doi.org/10.1038/s41558-026-02581-5

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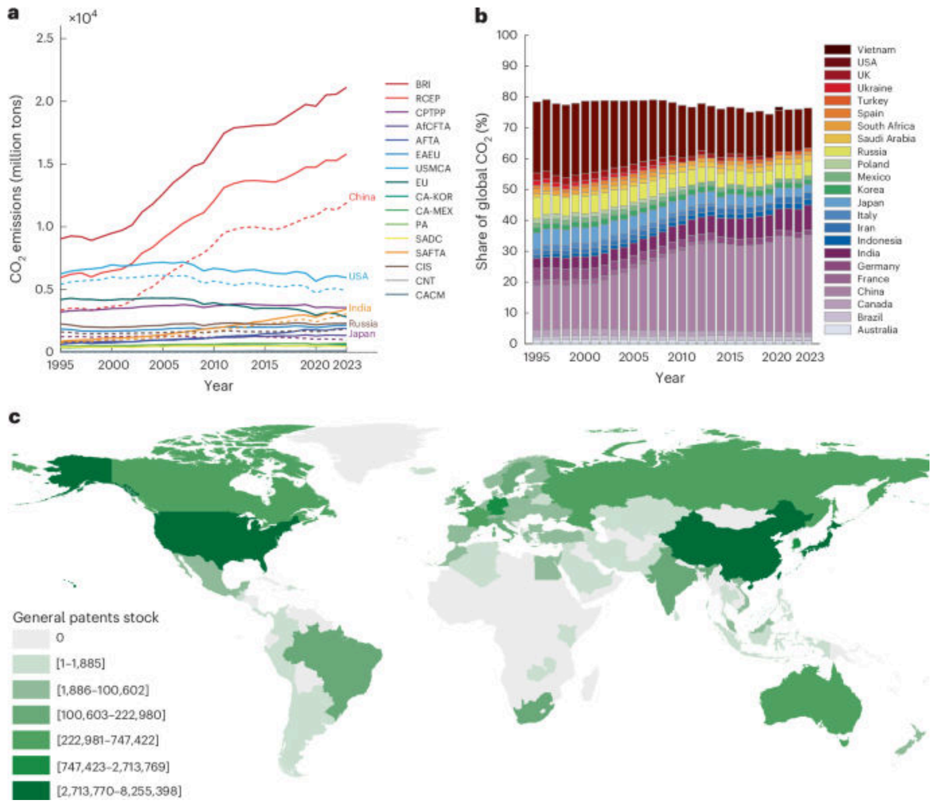
98. International trade reduces emissions through technology transfer led by key emitters

【国际贸易通过主要排放国主导的技术转让减少排放】

☐ Nature Climate Change | 2026-03-17 [Latest] | 重要性: 49 分

☐ Graphical Abstract:



☐ Abstract: Nature Climate Change, Published online: 17 March 2026; doi:10.1038/s41558-026-02595-zTechnology advancement is essential for climate action, yet the uneven distribution of technological progress across the world can slow mitigation. Through empirical and scenario analysis, researchers find that participating in trade agreements could enhance technological transfers and lead to emission reductions.

☐ 中文摘要:

技术进步对气候行动至关重要，但全球各地技术进步的不均衡分布可能减缓减缓行动。通过实证和情景分析，研究者发现主要排放国的清洁技术出口通过技术转让机制，对全球减排产生了显著的溢出效应。

☐ 深度解读:

本研究发表于 Nature Climate Change，提供了国际贸易与全球减排之间关系的实证证据，填补了贸易-技术-碳排放三者关系研究的空白。研究揭示了一个关键机制：中国等主要排放国通过出口太阳能板、新能源汽车等清洁技术产品，正在成为全球绿色技术扩散的重要驱动力。这一发现对国际气候谈判框架具有政策含义，也为一带一路绿色发展战略提供了科学支撑，表明贸易与气候目标之间可以形成正向协同而非必然冲突。

☐ DOI: https://doi.org/10.1038/s41558-026-02595-z

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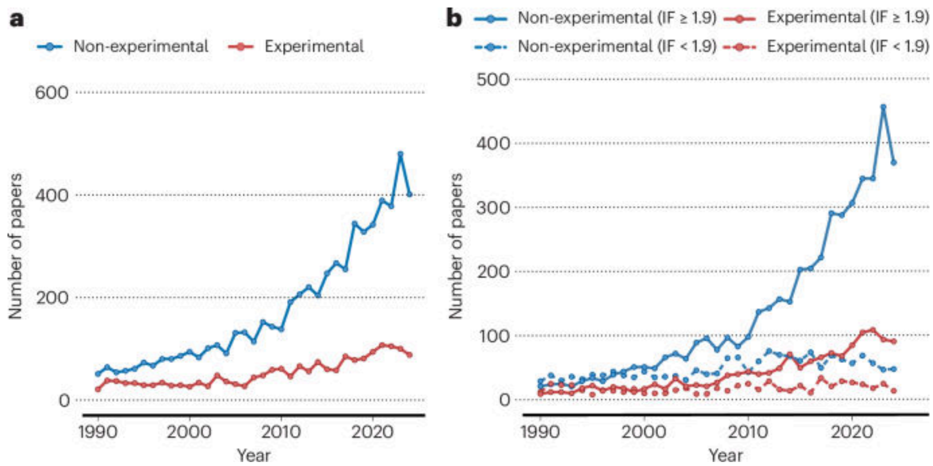
99. Increasing risks of post-experimental ecology

【后实验生态学的风险日益增加】

☐ Nature Climate Change | 2026-03-13 [Latest] | 重要性: 49 分

☐ Graphical Abstract:



☐ Abstract: Nature Climate Change, Published online: 13 March 2026; doi:10.1038/s41558-026-02583-3Increasing risks of post-experimental ecology

☐ 中文摘要:

本评论文章探讨了生态学实验研究中一种日益突出的风险：在气候变化背景下，某些生态实验的实施本身可能对生态系统造成无法逆转的干扰，引发关于实验设计和生态系统保护之间道德和方法论张力的讨论。

☐ 深度解读:

本文发表于 Nature Climate Change，引发了对生态学研究范式的深刻反思：在气候变化加速的背景下，一些大规模野外操控实验本身正在成为生态风险的来源，尤其是在脆弱生态系统（热带雨林、珊瑚礁）中进行的破坏性实验。研究呼吁生态学界建立更严格的实验伦理框架，平衡知识获取与生态系统保护的关系。这对中国正在扩张的野外生态学研究网络（CNERN 等）的实验设计规范和伦理审查制度建设具有重要启示。

☐ DOI: https://doi.org/10.1038/s41558-026-02583-3

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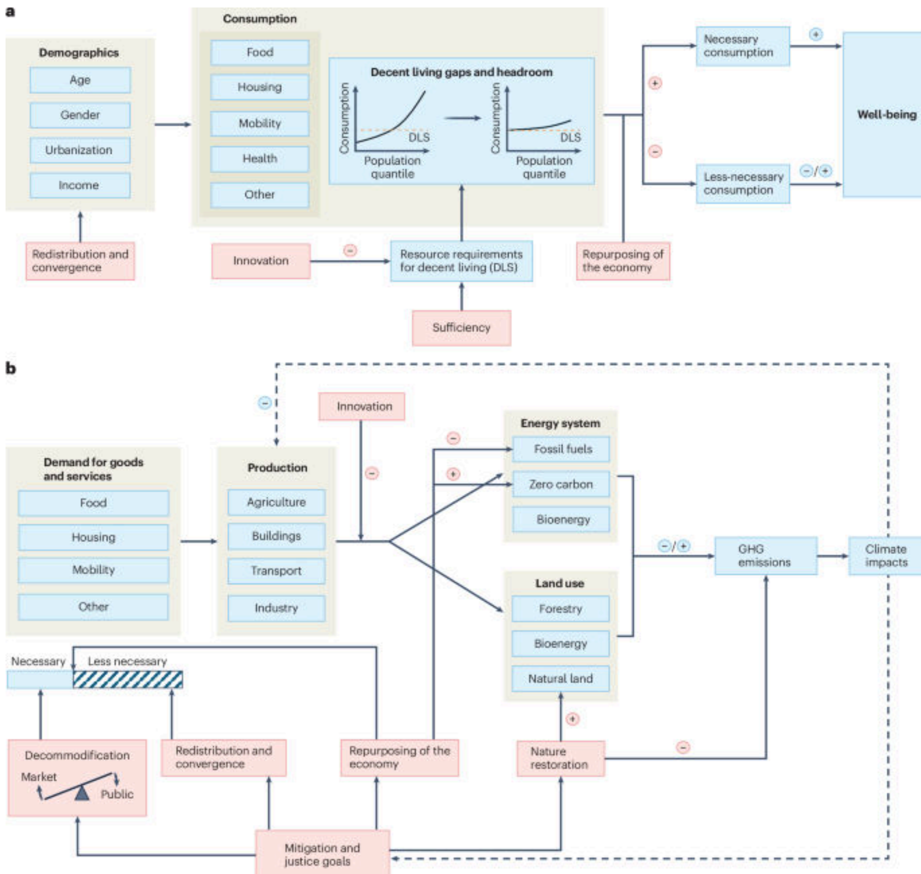
100. Principles for a post-growth scenario of ambitious mitigation and high human well-being

【高质量生活与雄心减排后增长情景的核心原则】

☐ Nature Climate Change | 2026-03-13 [Latest] | 重要性: 49 分

☐ Graphical Abstract:



☐ Abstract: Nature Climate Change, Published online: 13 March 2026; doi:10.1038/s41558-026-02580-6Post-growth scholarship seeks to address the limitations of growth-oriented mitigation scenarios by exploring the potential of profound socio-economic transformations. This Perspective synthesizes core principles for modelling post-growth futures.

☐ 中文摘要:

后增长学术研究通过探索深刻的社会经济转型潜力，试图解决以增长为导向的减缓情景的局限性。本综述综合了指导后增长情景建模的核心原则，为气候政策制定提供超越 GDP 增长导向的新范式框架。

☐ 深度解读:

本文发表于 Nature Climate Change，是后增长气候情景研究的重要综述，系统总结了在不依赖持续经济增长的前提下实现气候雄心与人类福祉双赢的原则框架。研究挑战了主流气候模型中经济增长与减排脱钩的基本假设，提出通过减少过度消费、公平分配资源等社会转型路径实现碳中和。这对理解中国生态文明建设与高质量发展的内在一致性具有理论意义，也为探索中国式现代化路径的全球气候贡献提供了新的分析视角。

☐ DOI: https://doi.org/10.1038/s41558-026-02580-6

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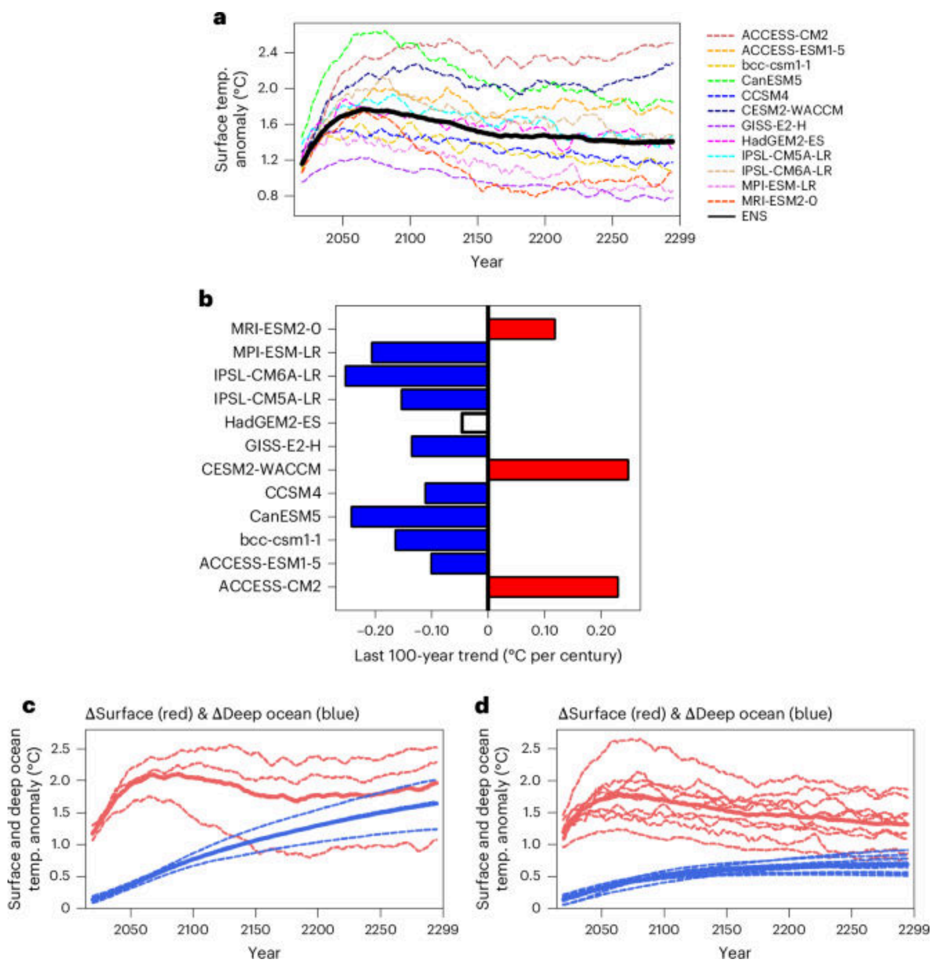
101. Deep ocean control of global temperature after net-zero emissions

【净零排放后深海对全球温度的调控作用】

☐ Nature Geoscience | 2026-03-16 [Latest] | 重要性: 44 分

☐ Graphical Abstract:



☐ Abstract: Nature Geoscience, Published online: 16 March 2026; doi:10.1038/s41561-026-01934-1When atmospheric carbon dioxide levels decline, increased ocean heat absorption would initially slow surface warming before warming the deep ocean, which ultimately raises global mean surface temperatures, according to future model simulations.

☐ 中文摘要:

当大气二氧化碳浓度下降时，海洋吸热量增加将首先减缓地表升温，但随后会加热深海，最终导致全球平均地表温度升高。这一深海热力学反馈机制揭示了实现净零排放后的气候演变路径。

☐ 深度解读:

本研究发表于 Nature Geoscience，揭示了净零排放达成后全球气候演变的一个关键但此前被低估的物理机制：深海热容量的延迟释放效应。研究表明即使人类实现净零排放，由于深海热容量巨大，地球表面温度仍可能在数十年至数百年内继续上升，这对气候目标的理解有重大修正意义。该发现提示实现净零排放并不意味着气候问题立即解决，还需要长期监测和可能的碳移除技术，对政策制定者具有重要启示。

☐ DOI: https://doi.org/10.1038/s41561-026-01934-1

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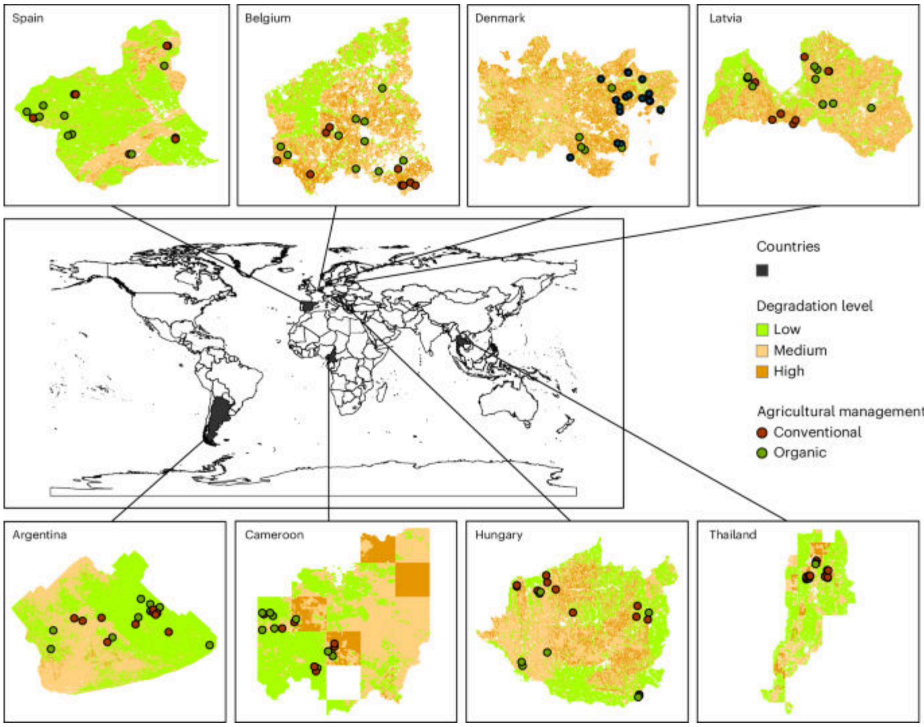
102. Optimizing biodiversity, multifunctionality and yield when transitioning to organic farming

【向有机农业转型时优化生物多样性、多功能性和产量】

☐ Nature Sustainability | 2026-03-16 [Latest] | 重要性: 40 分

☐ Graphical Abstract:



☐ Abstract: Nature Sustainability, Published online: 16 March 2026; doi:10.1038/s41893-026-01791-1Different agricultural management systems, for example, conventional versus organic, can have different benefits and challenges. Authors here examine the biodiversity, crop yield and ecosystem multifunctionality impacts of transitioning from conventional to organic agriculture across 179 global croplands.

☐ 中文摘要:

不同农业管理系统（如常规农业与有机农业）具有不同的效益和挑战。本研究考察了生物多样性、作物产量和生态系统多功能性在不同管理系统中的表现，探讨如何在向有机农业转型过程中优化这些目标间的权衡关系。

☐ 深度解读:

本研究发表于 Nature Sustainability，通过系统分析欧洲多个长期农业实验数据，揭示了有机农业转型中生物多样性、生态系统多功能性和作物产量之间的协同与权衡关系。研究发现通过优化农场景观多样性和管理实践组合，可以在产量损失最小的前提下实现生物多样性和生态功能的最大化提升。这一发现对中国推进耕地保护与绿色农业发展具有重要参考意义，特别是在实现昆明-蒙特利尔框架恢复目标过程中的农业用地管理策略选择上。

☐ DOI: https://doi.org/10.1038/s41893-026-01791-1

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103. Managing nitrogen for food and environment

【为粮食安全与环境保护管理氮素】

☐ Nature Geoscience | 2026-03-12 [Latest] | 重要性: 40 分

☐ Graphical Abstract:

☐ Abstract: Nature Geoscience, Published online: 12 March 2026; doi:10.1038/s41561-026-01947-wUnsustainable use of nitrogen fertilizers and ineffective control of nitrogen losses have resulted in various environmental issues. A key challenge for nitrogen management is to meet the food demands of a growing global population while reducing nitrogen pollution.

☐ 中文摘要:

氮肥的不可持续使用和对氮素损失的无效控制导致了各种环境问题。在满足全球不断增长的粮食需求的同时减少氮损失，是氮素管理的关键挑战。研究提出了兼顾粮食安全与环境保护的综合氮素管理框架。

☐ 深度解读:

本研究发表于 Nature Geoscience，提供了迄今最全面的全球氮素管理综合评估，揭示农业氮素管理面临的粮食安全与环境保护双重压力之间的根本矛盾。研究定量化了全球氮素流失的热点区域和主要途径，提出了基于氮足迹概念的精准管理策略。对中国尤为重要：中国是全球最大的氮肥消费国，过量施氮导致的水体富营养化、大气污染和土壤酸化问题严峻，本研究为中国化肥减量增效行动提供了系统性科学框架。

☐ DOI: https://doi.org/10.1038/s41561-026-01947-w

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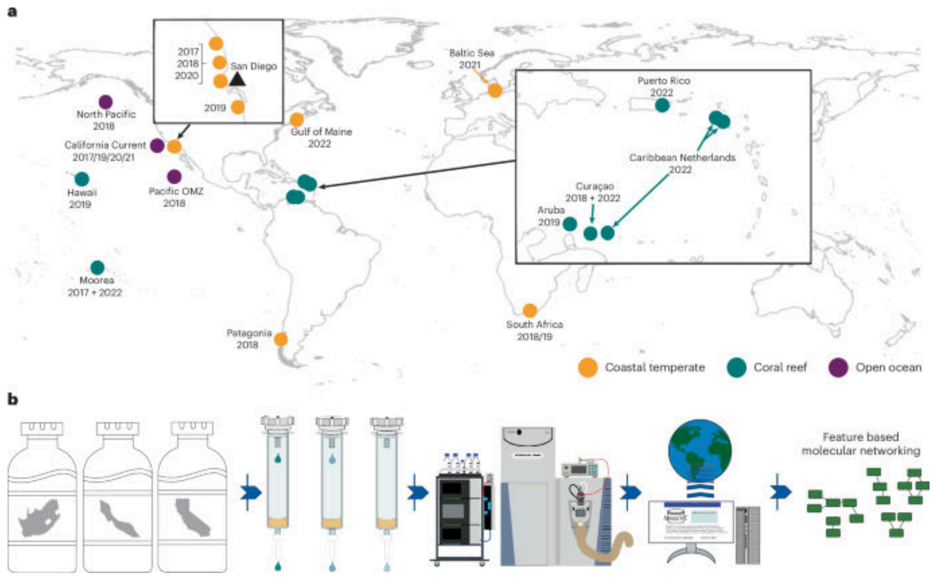
104. Widespread presence of anthropogenic compounds in marine dissolved organic matter

【海洋溶解有机物中人为化合物的广泛存在】

☐ Nature Geoscience | 2026-03-16 [Latest] | 重要性: 38 分

☐ Graphical Abstract:



☐ Abstract: Nature Geoscience, Published online: 16 March 2026; doi:10.1038/s41561-026-01928-zSynthetic chemicals can make up substantial proportions of dissolved organic matter in the ocean, with major differences in compound classes depending on the distance from coastlines, according to an analysis of seawater samples from around the globe.

☐ 中文摘要:

合成化学品可能占海洋中溶解有机物的相当比例，化合物类别的主要差异取决于距海岸线的距离。这一发现揭示了人类活动对全球海洋化学组成的深远影响。

☐ 深度解读:

本研究发表于 Nature Geoscience，首次在全球尺度上系统记录了人为合成化学品在海洋溶解有机物库中的广泛分布，包括持久性有机污染物、药品活性化合物和个人护理品等。研究利用高分辨质谱技术，揭示即使在远离人类活动的深海区域也检测到大量合成化学品，表明海洋已成为全球人为化学品的汇。这一发现对评估全球污染物的生态风险具有里程碑意义，对中国海洋污染治理政策和新污染物管控法规的实施提供了重要科学背景。

☐ DOI: https://doi.org/10.1038/s41561-026-01928-z

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105. Microbial upcycling of plastic waste to levodopa

【微生物将塑料废物升级利用生产左旋多巴】

☐ Nature Sustainability | 2026-03-16 [Latest] | 重要性: 36 分

☐ Abstract: Nature Sustainability, Published online: 16 March 2026; doi:10.1038/s41893-026-01785-zThe production of high-value chemicals can involve energy-intensive processes, necessitating sustainable production strategies. Here the authors present a circular bioeconomy approach, upcycling plastic waste through microbial conversion into levodopa, a medicine for Parkinson’s disease.

☐ 中文摘要:

高价值化学品的生产往往涉及能源密集型工艺，迫切需要可持续的生产策略。本研究提出一种循环生物经济方法，通过微生物代谢途径将塑料废物升级转化为左旋多巴，实现塑料污染的高值化利用。

☐ 深度解读:

本研究发表于 Nature Sustainability，展示了一条将塑料污染治理与高价值化学品生物制造相结合的创新路径，代表了循环生物经济的突破性进展。研究利用合成生物学工具改造微生物代谢网络，使其能以降解塑料产生的小分子为原料合成具有高药用价值的左旋多巴，实现废物变良药的范式转变。

这一成果为解决全球塑料污染问题提供了全新的高值化解决方案，对中国无废城市建设和合成生物学产业发展具有重要启示。

☐ DOI: https://doi.org/10.1038/s41893-026-01785-z

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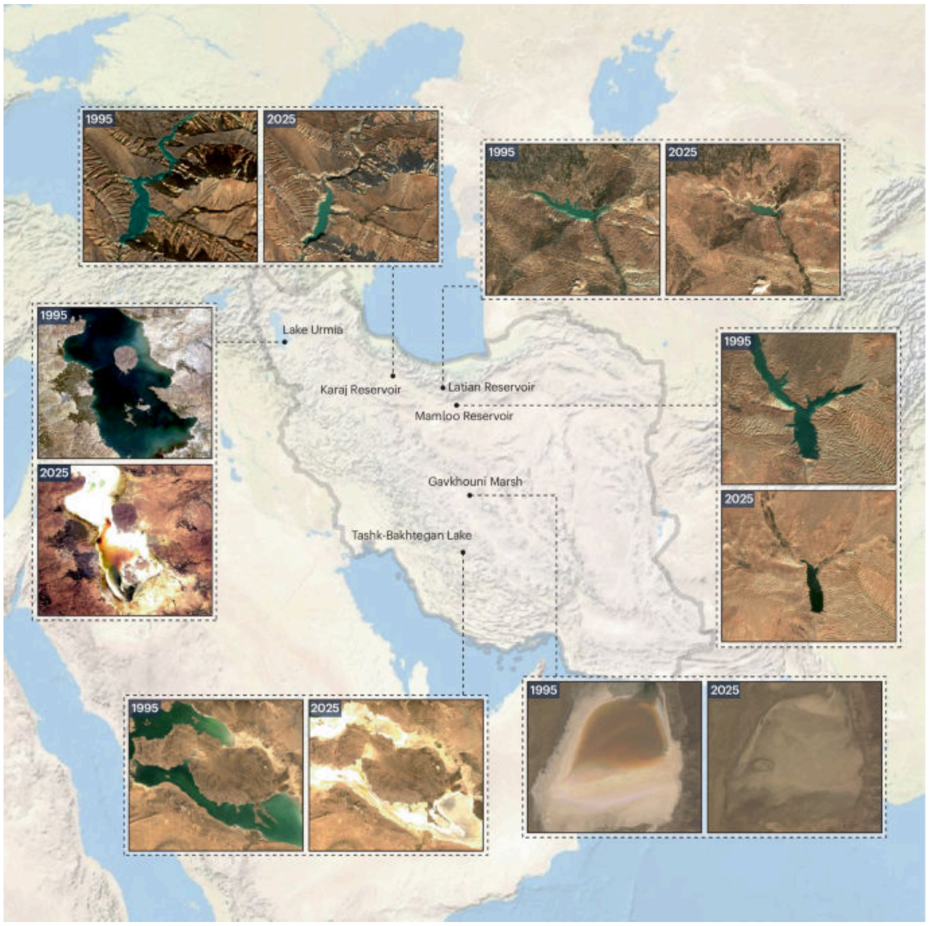
106. Iran’s policy priorities intensify water crisis

☐ Nature Sustainability | 2026-03-13 [Latest] | 重要性: 36 分

☐ Graphical Abstract:



☐ Abstract: Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01784-0The chronic water crisis in Iran stems from decades of water-intensive development, fragmented governance and national priorities that sidelined environmental protection. Without major governance reform and reprioritization, technical solutions alone cannot stop worsening water and environmental degradation.

☐ DOI: https://doi.org/10.1038/s41893-026-01784-0

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107. Early action for coastal communities

【对沿海社区采取早期行动】

☐ Nature Geoscience | 2026-03-12 [Latest] | 重要性: 36 分

☐ Graphical Abstract:

☐ Abstract: Nature Geoscience, Published online: 12 March 2026; doi:10.1038/s41561-026-01932-3Shifts in large-scale climate patterns are reshaping flood risk worldwide. Advances in modelling now offer the potential to provide early warnings and develop effective tools for managing rising coastal hazards.

☐ 中文摘要:

大尺度气候模式的转变正在重塑全球洪水风险格局。建模技术的进步为提供预警和开发有效应对沿海灾害工具提供了可能。本研究提出针对沿海社区的气候风险早期预警和适应行动框架。

☐ 深度解读:

本研究发表于 Nature Geoscience，利用最新一代高分辨率气候模式，评估了气候变化背景下全球沿海洪涝风险的时空演变格局，并提出基于预警系统的早期行动框架。研究特别关注气候模式转变（如 ENSO 频率变化、热带气旋路径移动）对沿海洪水风险的非线性影响。这一研究对中国长三角、珠三角等高密度沿海城市群的气候适应规划具有重要参考意义，对国家气候适应战略的制定和实施提供了科学支撑。

☐ DOI: https://doi.org/10.1038/s41561-026-01932-3

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108. [ASAP] Bacterial Transformation Products from Organophosphate Esters and Their Elevated Environmental Risks in Soil and Groundwater Surrounding a Chemical Industrial Park

☐ Environmental Science & Technology | Tue, 17 Mar 2026 12: [Latest] | 重要性: 36 分

☐ Abstract: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c13940

☐ DOI: https://doi.org/10.1021/acs.est.5c13940

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109. [ASAP] Unexpected Polymerization Pathway in the Carbocatalysts/Permanganate Processes for Water Decontamination

☐ Environmental Science & Technology | Tue, 17 Mar 2026 05: [Latest] | 重要性: 36 分

☐ Abstract: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c17609

☐ DOI: https://doi.org/10.1021/acs.est.5c17609

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110. [ASAP] Precipitation-Driven Shifts in Organic Sulfur Decomposition and Oxidation State along Rainfall Gradients

☐ Environmental Science & Technology | Mon, 16 Mar 2026 09: [Latest] | 重要性: 36 分

☐ Abstract: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c07788

☐ DOI: https://doi.org/10.1021/acs.est.5c07788

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111. Epigenotoxicity of Bisphenol A altered global and ten-eleven translocation-dependent DNA methylation in a non-model fish

☐ Science of the Total Environment [Latest] | 重要性: 36 分

☐ Abstract: Publication date: 20 April 2026Source: Science of The Total Environment, Volume 1026Author(s): Sourav Kundu, K. Abhilash Wodeyar, Subhamoy Dutta, Subhadeep Das Gupta, Basanta Kumar Das, Gunjan Karnatak, Pranaya Kumar Parida

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0048969726003463

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112. Effect of climate-driven childhood sleep erosion on potential regional economic inequality

☐ Nature Sustainability | 2026-03-13 [Latest] | 重要性: 34 分

☐ Abstract: Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01779-xHot climates erode children’s sleep, threatening cognitive development with likely socioeconomic consequences. A global analysis finds potential links between future sleep losses and trillions of dollars in economic costs that disproportionately burden poorer regions.

☐ DOI: https://doi.org/10.1038/s41893-026-01779-x

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113. The critical role of soil moisture in compound hazards

☐ Nature Geoscience | 2026-03-13 [Latest] | 重要性: 34 分

☐ Abstract: Nature Geoscience, Published online: 13 March 2026; doi:10.1038/s41561-026-01936-zSoil moisture drives critical land–atmosphere interactions and compound hazards. Closing gaps in data and models is key to integrating soil moisture into predictive frameworks for improved hazard forecasting.

☐ DOI: https://doi.org/10.1038/s41561-026-01936-z

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114. [ASAP] High Monoterpenoid Emissions from Scots Pine Litter Controlled by Moisture

☐ Environmental Science & Technology | Tue, 17 Mar 2026 12: [Latest] | 重要性: 34 分

☐ Abstract: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c17450

☐ DOI: https://doi.org/10.1021/acs.est.5c17450

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115. Source-dependent temporal toxicity and transcriptomic remodeling by urban ultrafine particles: Megacity - suburban comparison in aging olfactory-brain interface

☐ Science of the Total Environment [Latest] | 重要性: 34 分

☐ Abstract: Publication date: 20 April 2026Source: Science of The Total Environment, Volume 1026Author(s): Claire Fayad, Laura Mussalo, Aleksei Afonin, Riikka Lampinen, Henri Hakkarainen, Anna-Katharina Hensel, Karim Yahfoufi, Donya Behzadpour, Elina Penttilä, Anne M. Koivisto, Luciano Cascione, Sari Pennings, Santtu Mikkonen, Pasi Jalava, Katja M. Kanninen

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0048969726003220

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116. Ecoacoustics tools reveal effects of human-caused wildfires on anuran communities of the Yungas Andean Forest

☐ Science of the Total Environment [Latest] | 重要性: 34 分

☐ Abstract: Publication date: 20 April 2026Source: Science of The Total Environment, Volume 1026Author(s): Boullhesen Martin, Luypaert Thomas, Sousa-Lima Renata, Vaira Marcos, Akmentins Mauricio Sebastían

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0048969726003219

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117. Differences in microplastics in passerine feces across species, diet, and foraging location

☐ Science of the Total Environment [Latest] | 重要性: 34 分

☐ Abstract: Publication date: 20 April 2026Source: Science of The Total Environment, Volume 1026Author(s): Victoria Moreira, Jennifer J. Uehling, Alison Fetterman, Lisa Kiziuk, Michelle A. Eshleman, Abbie Ganas, Megan L. Fork

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0048969726003426

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118. Life cycle assessment of microwave torrefaction and pellet production from agri-forest residues

☐ Science of the Total Environment [Latest] | 重要性: 34 分

☐ Abstract: Publication date: 20 April 2026Source: Science of The Total Environment, Volume 1026Author(s): Obiora S. Agu, Lope G. Tabil, Edmund Mupondwa, Xue Li

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0048969726003621

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119. [ASAP] Using a Multi-Tracer Approach to Examine Perfluoroalkyl Substance Sources and Dietary Exposure Pathways in Pacific Bald Eagles

☐ Environmental Science & Technology | Tue, 17 Mar 2026 03: [Latest] | 重要性: 32 分

☐ Abstract: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c15624

☐ DOI: https://doi.org/10.1021/acs.est.5c15624

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120. High-resolution models better simulate historical snowpack declines in the Upper Colorado River Basin

☐ Environmental Research Letters | 2026-03-17T00:00:00Z [Latest] | 重要性: 25 分

☐ Abstract: The Colorado River supplies water to 40 million people, 4.5 million acres of irrigated land, seven US states, over two dozen federally recognized and sovereign tribes and Mexico. River discharge is strongly controlled by snowpack accumulation and melt in the headwater regions, making upper-basin snow dynamics a primary driver of water availability throughout the basin. Therefore, changes in snow water equivalent (SWE) are critical for water resources planning in the Upper Colorado River Basin (UCRB), yet there is considerable uncertainty even in historical SWE trends. We compare historical UCRB SWE trends across various gridded snow products, including new Weather Research and Forecasting simulations at 45, 9, and 3 km resolution, against snow telemetry station data. We find that SWE trend

☐ DOI: https://iopscience.iop.org/article/10.1088/1748-9326/ae4113

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121. Land use and land cover change intensified soil moisture drought: evidence from CMIP6-LUMIP

☐ Environmental Research Letters | 2026-03-17T00:00:00Z [Latest] | 重要性: 25 分

☐ Abstract: Soil moisture drought (SMD) critically constrains terrestrial ecosystems and agricultural productivity, yet the long-term contribution of historical land use and land cover change (LULCC) to global drought trends remains insufficiently quantified. We quantify the impacts of historical LULCC on SMD characteristics (event number, duration, severity, intensity) over 1901–2014 using seven offline land surface simulations from the CMIP6 Land Use Model Intercomparison Project (LUMIP). LULCC intensifies SMD over more than half of the global land area, with the strongest increases in mid-latitude regions, particularly in central North America. The drought-affected area expanded throughout the twentieth century, with drought events becoming increasingly prolonged and severe. Diagnostic analyses of

☐ DOI: https://iopscience.iop.org/article/10.1088/1748-9326/ae4ca5

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122. Tracking shifts in European drought hotspots

☐ Environmental Research Letters | 2026-03-17T00:00:00Z [Latest] | 重要性: 23 分

☐ Abstract: Drought is a complex hazard with far-reaching socio-economic and ecological impacts that are often inadequately captured by conventional indices based solely on hydroclimatic anomalies. To bridge this gap, this study develops novel impact-based combined drought indices (iCDIs) that directly relate hydroclimatic drivers to remotely sensed vegetation stress, here used as a proxy for observed agricultural impacts. The proposed framework employs a machine learning framework to optimize the selection of hydroclimatic predictors including precipitation, temperature, and soil moisture across distinct hydrological clusters of the European domain. This bottom-up machine learning-based approach outperforms traditional indices in reproducing observed vegetation responses across European river basins.

☐ DOI: https://iopscience.iop.org/article/10.1088/1748-9326/ae4ca7

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123. The effects of self-governance arrangements on the climate change resilience of small-scale fisheries in Mexico

☐ Global Environmental Change [Latest] | 重要性: 21 分

☐ Abstract: Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Xochitl Édua Elías Ilosvay, Sarah Elkin, Sebastian C.A. Ferse, Eréndira Aceves-Bueno, Javier Tovar-Ávila, Jhosafat Rentería-Bravo, Irving Alexis Medina Santiago, Jorge García Molinos, Elena Ojea

☐ DOI: https://www.sciencedirect.com/science/article/pii/S095937802600021X

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124. A study of private sector engagement in climate change mitigation and adaptation activities within pacific Small Island Developing States

☐ Climatic Change | 2026-03-14 [Latest] | 重要性: 19 分

☐ DOI: https://link.springer.com/article/10.1007/s10584-026-04147-y

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125. Stakeholder engagement in climate change mitigation: A focus on Tanzania’s renewable energy sector

☐ Climatic Change | 2026-03-06 [Latest] | 重要性: 19 分

☐ DOI: https://link.springer.com/article/10.1007/s10584-026-04132-5

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126. Unravelling the impact of changing rainy seasons on rice production in Southeast Asia: a climate change high-resolution perspective

☐ Climatic Change | 2026-03-16 [Latest] | 重要性: 17 分

☐ DOI: https://link.springer.com/article/10.1007/s10584-026-04136-1

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127. Decadal prediction system based on the INM RAS climate model

☐ Climate Dynamics | 2026-03-16 [Latest] | 重要性: 12 分

☐ Abstract: Based on the INMCM5 coupled general circulation model, the decadal climate prediction system has been developed. The system provides a complete cycle of decadal predictions from preparation of initial conditions to prognostic ensemble calculations and analysis of the results. The system has been evaluated for the years 1960–2020. The quality of INMCM5 hindcasts for annual and 5-year mean global and regional near-surface air temperature, precipitation, and sea level pressure as well as timeseries of key climate indices is on par with the mean level of the World Meteorological Organization Lead Centre for Annual-to-Decadal Climate Prediction (WMO LC-ADCP) models. Besides estimates recommended by the WMO LC-ADCP, an assessment of the East Atlantic-West Russian Oscillation, Western Pacific Osc

☐ DOI: https://link.springer.com/article/10.1007/s00382-026-08047-w

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128. A global perspective: quantifying disturbance and reclamation of surface coal mining through remote sensing innovations

☐ Global Environmental Change [Latest] | 重要性: 9 分

☐ Abstract: Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Zhen Yang, Wenrui Tian, Tingting He, He Ren, Xingdong Wang, Huawei Jiang, Like Zhao, Yanchuang Zhao, Yuwei Chen

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0959378026000270

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129. Seasonal forecasting using the GenCast probabilistic machine learning model

☐ Climate Dynamics | 2026-03-16 [Latest] | 重要性: 7 分

☐ DOI: https://link.springer.com/article/10.1007/s00382-026-08077-4

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130. Structural compactness governs the environmental fate of polystyrene Nanoplastics: Reaggregation mechanisms in laboratory-scale aquatic systems

☐ Environmental Pollution [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Ji-Won Son, Soobin Yang, Seonho Lee, Hansang Lee, Seoyoung Park, Yoon Myung, Changwoo Kim

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0269749126003222

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131. Artificial sweetener acesulfame induces oxidative stress, neurotoxicity, and glycolipid metabolic disruption in zebrafish (Danio rerio)

☐ Environmental Pollution [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Xue Liu, Qian Wang, Huijuan Lv, BaoYan Mu, Yali Zhou, Xianxu Li, Min Ding, Nan Jiang, Jun Wang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0269749126003210

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132. Bisphenol analogs affected hepatic development and disrupted lipid homeostasis in zebrafish larvae

☐ Environmental Pollution [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Yao Zheng, Ying Huang, Jiajia Li, Liyan Teng, Yanyang Xu, Lilai Yuan, Wenbo Yang, Jing Qiu, Yongzhong Qian, Tingting Chai, Xiyan Mu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S026974912600299X

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133. High-molecular-weight species dominate the leaf soluble mercury pool: Rubisco as the primary binding protein

☐ Environmental Pollution [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Hui Tao, Yuhan Fan, Yanwei Liu, Yingying Guo, Xingwang Hou, Jiyan Liu, Yongguang Yin, Yong Cai, Guibin Jiang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0269749126003209

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134. Integrating data driven approaches and geospatial modeling to identify environmental hot spots and guide targeted restoration efforts

☐ Journal of Environmental Management [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 April 2026Source: Journal of Environmental Management, Volume 404Author(s): Subhasis Giri, Richard G. Lathrop, Zeyuan Qiu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0301479726007875

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135. Spatial patterns of environmental injustice in social vulnerability and ambient dust levels across Australia

☐ Environmental Research Letters | 2026-03-17T00:00:00Z [Latest] | 重要性: 6 分

☐ Abstract: Ambient dust exposure has been associated with several adverse health outcomes. Unlike ambient air pollution, the vulnerability of subpopulations to ambient dust has not yet been explored. As the driest inhabited continent, Australia provides a natural laboratory for studying large-scale dust exposure. This study aimed to examine environmental injustice in the ambient dust of ☐10 μm particle size and vulnerability of subpopulations in Australia, which has not been studied previously. A nationwide cross-sectional design was employed by linking a highly spatially resolved census-derived composite measure of social vulnerability index (a wide range of factors measuring susceptibility, and a more complex capacity of individuals and society to cope with hazards and damage) to 2021 annual mean a

☐ DOI: https://iopscience.iop.org/article/10.1088/1748-9326/ae4d5d

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136. Israel-Gaza conflict carbon emissions exceeded 30 million tons

☐ One Earth [Latest] | 重要性: 4 分

☐ Abstract: Publication date: Available online 11 March 2026Source: One EarthAuthor(s): Benjamin Neimark, Frederick Otu-Larbi, Reuben Tete Larbi, Patrick Bigger, Linsey Cottrell, Lennard de Klerk, Mykola Shlapak

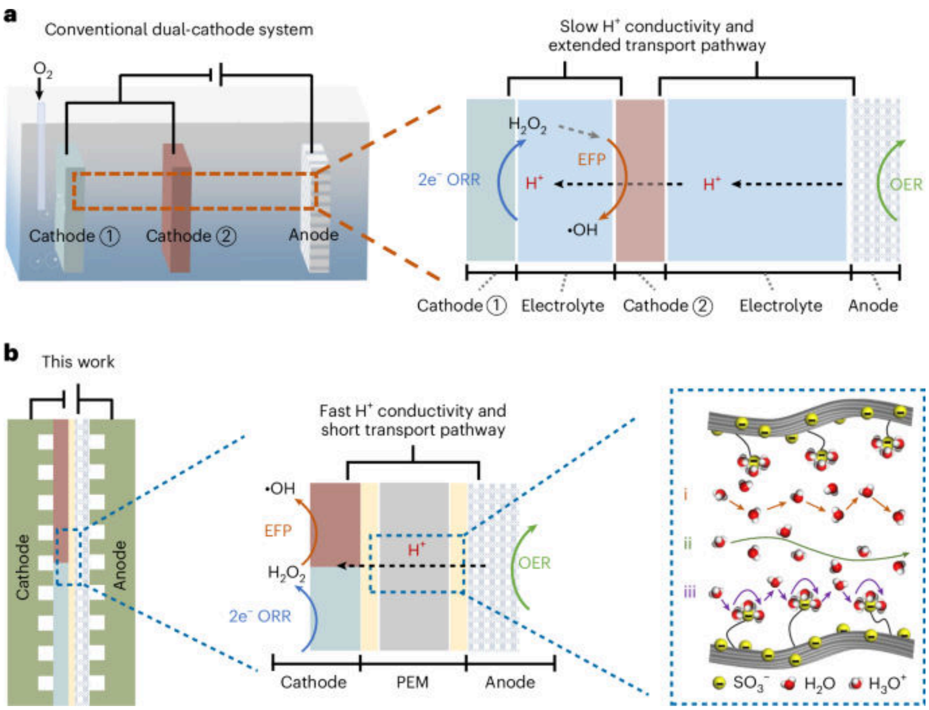
☐ DOI: https://www.sciencedirect.com/science/article/pii/S2590332226000497

【海洋与水生】

137. Molecular oxygen cascade reduction to •OH via coplanar dual-electrocatalytic zone achieving electrolyte-free water purification

☐ Nature Water | 2026-03-10 [Latest] | 重要性: 36 分

☐ Graphical Abstract:



☐ Abstract: Nature Water, Published online: 10 March 2026; doi:10.1038/s44221-026-00606-zThis study presents an electrolyte-free electrochemical system that leverages dual electrocatalytic zones to effectively remove a wide range of recalcitrant pollutants across various water matrices, including low-conductivity organic wastewater.

☐ DOI: https://doi.org/10.1038/s44221-026-00606-z

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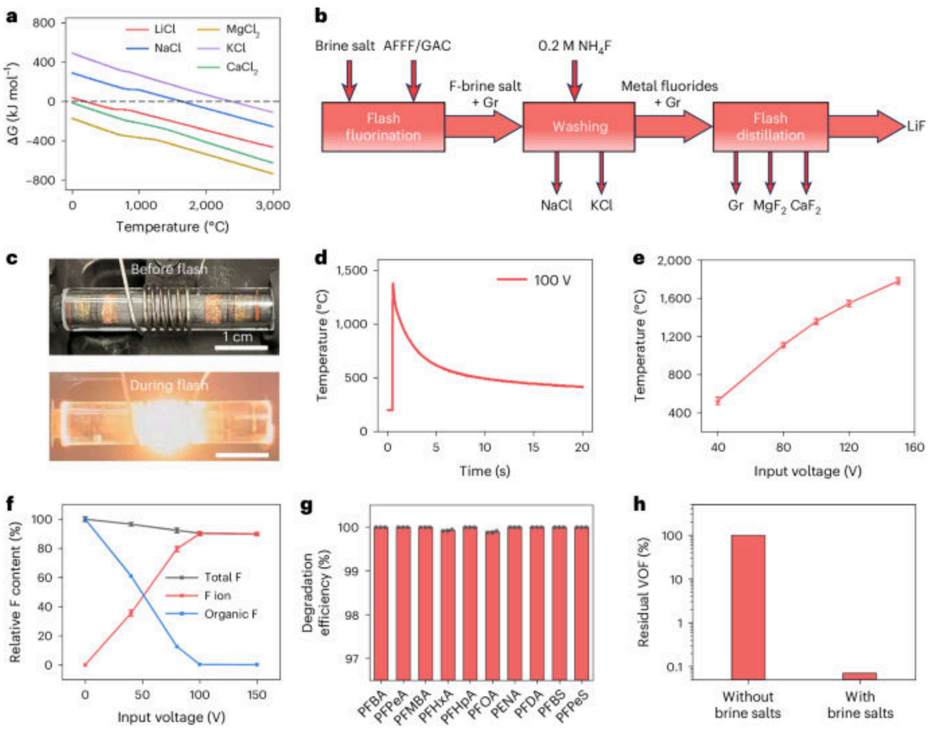
138. Waste per- and polyfluoroalkyl substance-assisted flash fluorination for lithium recovery from brine

☐ Nature Water | 2026-03-10 [Latest] | 重要性: 36 分

☐ Graphical Abstract:



☐ Abstract: Nature Water, Published online: 10 March 2026; doi:10.1038/s44221-026-00593-1An electrothermal fluorination methodology has been developed to transform granular activated carbon-sorbed aqueous film-forming foam waste into graphene and a fluorinating reagent, subsequently enabling lithium recovery from brine sources.

☐ DOI: https://doi.org/10.1038/s44221-026-00593-1

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139. Fishing vessels recording a two-decade dataset to model the spatio-temporal distribution of anchovy in the Adriatic Sea: a preliminary exploration

☐ Estuarine Coastal and Shelf Science [Latest] | 重要性: 34 分

☐ Abstract: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Enrico Cecapolli, Lorenzo Zacchetti, Filippo Domenichetti, Fabrizio Moro, Alberto Santojanni, Andrea Belardinelli, Pierluigi Penna, Michela Martinelli

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0272771426001174

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140. Numerical investigation of probabilistic models for wind–wave misalignment during historical typhoons

☐ Estuarine Coastal and Shelf Science [Latest] | 重要性: 32 分

☐ Abstract: Publication date: 15 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 335Author(s): Kai Wei, Yanchao He, Haoyu Li, Zhikai Li, Qingshan Yang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0272771426001149

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141. Global seasonal distribution of low-frequency ocean soundscapes

☐ Estuarine Coastal and Shelf Science [Latest] | 重要性: 32 分

☐ Abstract: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Junhao Zhang, Jiancheng Yu, Jie Sun, Shilong Li, Qi Zhang, Yu Tian, Xiaolong Yu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0272771426001022

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142. Meta analysis of Manning's coefficient for saltmarsh and mangroves to support broad scale assessment

☐ Estuarine Coastal and Shelf Science [Latest] | 重要性: 32 分

☐ Abstract: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Syed Shamsil Arefin, Robert J. Nicholls, Stefanie Nolte, Jack Heslop

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0272771426000867

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143. Interfacial interaction-driven rapid capture and on-site analysis of nano- and microplastics enabled by multifunctional magnetic adsorbent

☐ Nature Water | 2026-03-13 [Latest] | 重要性: 32 分

☐ Abstract: Nature Water, Published online: 13 March 2026; doi:10.1038/s44221-026-00610-3A multifunctional adsorbent, composed of copper-doped polydopamine-functionalized magnetic silica, enables the simultaneous removal and on-site detection of label-free nano- and microplastics.

☐ DOI: https://doi.org/10.1038/s44221-026-00610-3

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144. Steering the nitrate electroreduction pathway via nanoconfinement-induced hydrogen-bond network regulation

☐ Nature Water | 2026-03-11 [Latest] | 重要性: 32 分

☐ Abstract: Nature Water, Published online: 11 March 2026; doi:10.1038/s44221-026-00600-5Electrochemical nitrate reduction can simultaneously remove pollutants and produce ammonia but is often limited by hydrogen side reactions. This study shows that nanoconfined CuCo catalysts enable a proton-coupled electron transfer pathway that delivers highly selective, efficient and stable ammonia synthesis.

☐ DOI: https://doi.org/10.1038/s44221-026-00600-5

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145. Water scientists must become agenda-setters

☐ Nature Water | 2026-03-09 [Latest] | 重要性: 32 分

☐ Abstract: Nature Water, Published online: 09 March 2026; doi:10.1038/s44221-026-00613-0The global water agenda is outdated and narrow and is framed mainly as a downstream impact sector. Scientists must step up to help the world recognize water as an opportunity sector and to design a bolder water agenda.

☐ DOI: https://doi.org/10.1038/s44221-026-00613-0

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146. A framework to inform economic valuation of non-use benefits from coral reef intervention efforts

☐ Coral Reefs | 2026-03-17 [Latest] | 重要性: 31 分

☐ Abstract: Healthy coral reefs hold significant value to society, not just from industries such as tourism or fishing, but by their very existence. These so-called ʻnon-use benefitsʼare increasingly endangered as the health of coral reefs faces escalating threats from climate change. Yet, current observation and modelling tools that help guide reef intervention efforts typically do not provide guidance on how the impact of management actions on reef ecology flows through to changes in non-use benefits. Here, we demonstrate how ecological metrics, as measured by existing reef observation or modelling tools, can be translated to an economic valuation of non-use benefits from conservation actions on Australiaʼs Great Barrier Reef (GBR). Using our framework, we find that an additional square kilometre o

☐ DOI: https://link.springer.com/article/10.1007/s00338-026-02844-9

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147. Exploring salt wedge dynamics and nature-inspired mitigation measures in the lower Mississippi River

☐ Estuarine Coastal and Shelf Science [Latest] | 重要性: 30 分

☐ Abstract: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Ahmed Khalifa, Ehab Meselhe, Claire Kemick, Kelin Hu, Mead Allison

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0272771426001095

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148. Hydroregime-dependent shifts in zooplankton hatching under increasing temperatures in Mediterranean temporary ponds: an experimental approach

☐ Hydrobiologia | 2026-03-13 [Latest] | 重要性: 27 分

☐ Abstract: Zooplankton hatching is triggered by a series of environmental cues, and changes in these cues may alter hatching patterns. Our main objective was to assess how a 4 °C temperature increase influences timing and abundance of zooplankton hatching in temporary ponds from two Mediterranean-climate regions, Chile and Spain. For this, we conducted hatching experiments using microcosms with pond sediments. For each region, 15 ponds were selected, representing three hydroperiod types: semipermanent, seasonal, and ephemeral. Microcosms were incubated under two temperatures: current autumn conditions for Mediterranean-climate regions (15 to 23 °C) and a projected 4 °C increase. Warming significantly influenced timing and abundance of hatching of specific taxa across hydroregimes. Under warmer condit

☐ DOI: https://link.springer.com/article/10.1007/s10750-026-06183-2

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149. Evaluating satellite and modeled lake surface water temperature across the contiguous United States

☐ Hydrobiologia | 2026-03-11 [Latest] | 重要性: 26 分

☐ Abstract: We developed a model to predict surface water temperature across U.S. lakes using satellite remote sensing and in situ observations to enhance cyanobacterial harmful algal bloom (cyanoHAB) forecasting. The study focused on Sentinel-3 Ocean and Land Colour Instrument (OLCI) sensor resolved lakes. We developed random forest models using both Landsat-derived and in-situ-measured surface water temperature. Landsat models offered

broad spatial and temporal coverage of all OLCI resolved lakes, but they were sensitive to cloud cover and required filtering to minimize error. In contrast, the in situ model represented fewer OLCI resolved lakes, but yielded lower mean absolute error and bias. The models predicted lake surface temperature across the entire calendar year, with best performance (RMSEap

☐ DOI: https://link.springer.com/article/10.1007/s10750-026-06178-z

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150. Outcomes of the fourth global coral bleaching (2023–2024) in the Maldives

☐ Coral Reefs | 2026-03-13 [Latest] | 重要性: 22 分

☐ Abstract: The fourth global coral bleaching event (2023 – 2024) was the most extensive and intense ever recorded, driven by unprecedented marine heatwaves. In the Maldives, one of the largest atoll systems in the Indian Ocean, the 2024 thermal anomaly reached record sea surface temperatures (31.5 °C) and degree heating weeks exceeding 9 °C-weeks, indicating cumulative stress levels above the mortality threshold for most coral taxa. We assessed the ecological impacts of this event across 18 reefs in central and southern atolls, comparing benthic community structure before and after bleaching.In the central atolls, live hard coral cover dropped by > 40% on average; the highest mortality (57%) occurred on a lagoon reef in Ari Atoll and disproportionately impacted branching and tabular Acropora. These los

☐ DOI: https://link.springer.com/article/10.1007/s00338-026-02850-x

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151. Climate change linked to infertility in well-known shark species

☐ Journal of Fish Biology | Thu, 12 Mar 2026 08: [Latest] | 重要性: 21 分

☐ Abstract: Journal of Fish Biology, EarlyView.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jfb.70404

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152. Flexing machine learning muscle: predicting lake calcium to prioritize invasive mussel monitoring in Alaska

☐ Hydrobiologia | 2026-03-11 [Latest] | 重要性: 21 分

☐ Abstract: Invasive zebra (Dreissena polymorpha) and quagga (Dreissena rostiformis bugensis) mussels (ZQM) are spreading across western North America, but are not yet known to be established in Alaska. If they invaded Alaska, ZQM could disrupt freshwater ecosystems supporting some of the world's most productive wild salmon fisheries. This study examined a key factor limiting ZQM invasion risk, the concentration of dissolved ionic calcium (Ca^{2+}), by synthesizing existing water quality data from published and unpublished sources and collecting new samples to fill data gaps for important salmon-bearing lakes across Alaska. We assembled a geospatial dataset including observed dissolved Ca^{2+} concentrations for 1406 lakes and used a randomForest model to predict Ca^{2+} concentrations for 1182 additional lake

☐ DOI: https://link.springer.com/article/10.1007/s10750-026-06127-w

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153. Deep Learning Methods for detecting marine vertebrates and invertebrates from satellite, aerial, and underwater imagery: A review

☐ Deep Sea Research I [Latest] | 重要性: 18 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Arthur Benard, Hervé Nikue Amassah, Guillaume Feuilloley

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967063726000312

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154. Stream and reservoir habitat effects on invasive crayfish behavior and injury

☐ Hydrobiologia | 2026-03-17 [Latest] | 重要性: 18 分

☐ Abstract: Invasive crayfish are common and impactful in freshwater ecosystems. Many invasive species are introduced into reservoirs, and these human-modified ecosystems may alter animals'behavioral traits and impacts on native species. We collected invasive virile crayfish (Faxonius virilis) and invasive rusty crayfish (Faxonius rusticus) from streams and reservoirs in Vermont, USA, recorded injury rates, and conducted on-site behavioral assays to measure boldness and exploratory behavior. We wanted to know if habitat type was associated with behaviors that could influence crayfish invasion success or impacts on native species. Our findings did not indicate that behavior was influenced by habitat type (e.g., stream or reservoir), but did indicate behavioral differences across sites and correlations

☐ DOI: https://link.springer.com/article/10.1007/s10750-026-06180-5

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155. Taxonomy and Functional Traits Modulate the Environmental Niche Dimensions of Crustacean Zooplankton Communities in Submerged Aquatic Vegetation Beds

☐ Freshwater Biology | Sun, 08 Mar 2026 06: [Latest] | 重要性: 14 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/fwb.70193

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156. Cryptic coral reef microhabitats and the functional role of brittle stars in biogenic sediment generation

☐ Coral Reefs | 2026-03-17 [Latest] | 重要性: 14 分

☐ Abstract: Recent studies have highlighted the ecological importance of cryptic habitat space on coral reefs. These microhabitats, ranging from framework cavities to intra-branch space within marine macroalgae, can harbour diverse and ecologically relevant taxa. Here we show that brittle stars, ubiquitous across such cryptic habitats (reported densities exceeding 100 ind. m^{-2}), represent an important, but previously neglected, source of skeletal sediment on reefs. Reef-derived carbonate sediment is generated by a range of reef biota and represents a critical geo-ecological process, contributing to reef structural development and vertical accretion via framework infilling, and the development and maintenance of lagoonal and shoreline habitats. Our sediment production estimates are based on skeletal ca

☐ DOI: https://link.springer.com/article/10.1007/s00338-026-02847-6

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157. Multiple spatial scales and habitat composition drive wrasse distributions on the Great Barrier Reef

☐ Coral Reefs | 2026-03-16 [Latest] | 重要性: 14 分

☐ Abstract: Wrasses (family Labridae) occupy a wide range of ecological niches, and although they are ubiquitous, their partitioning over large spatial and temporal scales is poorly understood. In this study, we surveyed the reef slope of 71 reefs between 9°S and 24°S along the Great Barrier Reef (GBR) to examine wrasse assemblages and benthic habitat associations. Seventy-two wrasse species from 24 genera were recorded. Most were widely distributed but relatively rare, with only a few exhibiting particularly high abundances or restricted distributions. Twenty-seven species accounted for 97% of the total wrasse density, and Cirrhilabrus punctatus was the single most abundant species. Distinct wrasse communities were observed on inner and outer shelf reefs, while mid-shelf reefs had high species overla

☐ DOI: https://link.springer.com/article/10.1007/s00338-026-02845-8

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158. The evolution, development, and genetic basis of cichlid hyoid shape reveals novel bone features and candidate genes

☐ Hydrobiologia | 2026-03-10 [Latest] | 重要性: 12 分

☐ Abstract: Among teleosts, the hyoid bone is critical for feeding, but the causes and consequences of hyoid shape variation are poorly understood relative to other elements in the head. We leverage the experimental tractability and evolutionary richness of East African cichlids to fill this knowledge gap. We first quantified hyoid shape variation in an ecologically diverse set of species, as well as in an F5 hybrid population derived from crossing two Lake Malawi cichlid species. We found that hyoid shape variation was both broad and nuanced, and only weakly predicted by foraging ecology. We then analyzed growth dynamics of the hyoid, which revealed a unique feature of the bone that differed between species—a series of interdigitating sutures that spanned the growth plate to connect the lateral (i.e.

☐ DOI: https://link.springer.com/article/10.1007/s10750-026-06172-5

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159. Reconstructing missing marine CDOM profiles using environmental covariates and graph-based deep learning

☐ Deep Sea Research I [Latest] | 重要性: 11 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Beibei Xie, Xinhui Liu, Fanzhi Meng, Di Zhang, Kailong Yuan

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967063726000300

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160. Multibeam seabed bathymetry of the Long-Term Ecological Research observatory HAUSGARTEN (Fram Strait)

☐ Deep Sea Research II [Latest] | 重要性: 11 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Boris Dorschel, Simon Dreutter, Lilian Boehringer

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967064526000305

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161. A physics-based framework for remote sensing inversion of fluorescent dissolved organic matter: incorporating fluorescence as an inelastic source term

☐ Water Research [Latest] | 重要性: 11 分

☐ Abstract: Publication date: 1 June 2026Source: Water Research, Volume 297Author(s): Ruiwu Zhang, Ruru Deng, Jun Ying

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0043135426004392

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162. Rethinking trophic transfer efficiency in freshwater ecosystems: Current understanding and knowledge gaps

☐ Limnology and Oceanography | Wed, 11 Mar 2026 06: [Latest] | 重要性: 10 分

☐ DOI: https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70343

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163. Fine-Scale Genomic Divergence in the Amazonian Pirarucu (Arapaima gigas) Highlights the Need for Local Management Strategies

☐ Freshwater Biology | Sun, 15 Mar 2026 20: [Latest] | 重要性: 9 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/fwb.70187

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164. Identifying Purse Seine Vessel Operations Through Machine Learning Models for Better Spatial Fishing Effort Estimates

☐ Fish and Fisheries | Fri, 13 Mar 2026 21: [Latest] | 重要性: 9 分

☐ Abstract: Fish and Fisheries, EarlyView.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/faf.70078

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165. Effects of microplastics on carbon and nitrogen cycling in coastal wetlands: A critical meta-analysis

☐ Water Research [Latest] | 重要性: 8 分

☐ Abstract: Publication date: 1 June 2026Source: Water Research, Volume 297Author(s): Feng Yuan, Hongyu Chen, Yue Xue, Yongcheng Ding, Qiujun An, Zhian Hu, Yifei Qiu, Jianguo Tao, Jin Luo, Guanghe Fu, Teng Wang, Xinqing Zou

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0043135426004458

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166. Unraveling the effects of wet and dry areas on carbon emissions in an intermittent Mediterranean stream

☐ Limnology and Oceanography | Thu, 12 Mar 2026 03: [Latest] | 重要性: 6 分

☐ DOI: https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70348

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167. Fecal pellet packaging enhances marine carbon sequestration

☐ Limnology and Oceanography | Wed, 04 Mar 2026 00: [Latest] | 重要性: 6 分

☐ DOI: https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70328

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168. Summertime methane and carbon dioxide emission rates and associated variables from a national-scale survey of 146 reservoirs in the United States

☐ Limnology and Oceanography Letters | Fri, 13 Mar 2026 06: [Latest] | 重要性: 6 分

☐ DOI: https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70080

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169. Aquatic reflectance derived from Sentinel-2 Multispectral Imager data for inland waters in the conterminous United States

☐ Limnology and Oceanography Letters | Sun, 22 Feb 2026 00: [Latest] | 重要性: 6 分

☐ DOI: https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70112

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170. Interactive Effects of Drought and Shading on Boreal Stream Metabolism and Algal Biomass in Experimental Mesocosms

☐ Freshwater Biology | Fri, 13 Mar 2026 06: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/fwb.70164

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171. Disentangling the Components Shaping Aquatic Richness Over Spatial Scales in Extreme Climatic Events

☐ Freshwater Biology | Thu, 12 Mar 2026 23: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/fwb.70194

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172. Harvesting that Preserves Large Fish Might Help Mitigate the Worst Impacts of Warming

☐ Fish and Fisheries | Mon, 16 Mar 2026 22: [Latest] | 重要性: 6 分

☐ Abstract: Fish and Fisheries, EarlyView.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/faf.70070

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173. A haplotype-specific PCR strategy for rapid and cost-effective identification of redfish species (genus Sebastes) in the North Atlantic and Arctic oceans

☐ Journal of Fish Biology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 6 分

☐ Abstract: Journal of Fish Biology, EarlyView.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jfb.70396

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174. Optical tracing of dissolved organic matter in the South Atlantic Ocean: linking water masses and biogeochemical processes

☐ Deep Sea Research I [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Clément Fériot, Céline Guéguen

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967063726000294

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175. Unlocking deep sea cold seep biodiversity: Leveraging environmental DNA for macrofauna diversity estimation in Haima cold seep

☐ Deep Sea Research I [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Gaoyou Yao, Hua Zhang, Wenguang Liu, Maoxian He

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967063726000282

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176. Drivers of benthic megafauna community dynamics in the Fram Strait, Arctic Ocean: Insights from the HAUSGARTEN observatory (2016-2021)

☐ Deep Sea Research II [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Lilian Boehringer, Melanie Bergmann, Christiane Hasemann, James Taylor, Jennifer Dannheim

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967064526000329

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177. Monitoring export fluxes to detect seasonal and interannual changes in the pelagic ecosystem of the northern Bering Sea

☐ Deep Sea Research II [Latest] | 重要性: 6 分

☐ Abstract: Publication date: April 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 226Author(s): Catherine Lalande, Phyllis J. Stabeno, Calvin W. Mordy, Shaun W. Bell, Chelsea W. Koch, Eun Jin Yang, Jinyoung Jung, Juyoung Son, Jee-Hoon Kim, Karen E. Frey, Jacqueline M. Grebmeier

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0967064526000226

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178. From food to sewer: Dietary inputs support the temporal and population-wide stability of pepper mild mottle virus in wastewater

☐ Water Research [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 1 June 2026Source: Water Research, Volume 297Author(s): Sofia Persson, Javier Edo Varg, Sarah Coker, Fanny Persson, Filip Petrini, Israa Dafalla, Lauren Davies, Polina Vaulina, Monika Vilemova, Nahla Mohamed, Maja Malmberg, Anna J. Székely

☐ DOI: https://www.sciencedirect.com/science/article/pii/S004313542600312X

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179. Irrigation Monitoring With Cosmic-Ray Neutron Sensors: Unraveling Field Experiments With Neutron Transport Simulations

☐ Water Resources Research | Sat, 14 Mar 2026 00: [Latest] | 重要性: 6 分

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR043105

【保护与生物多样性】

180. Contrasting vulnerability profiles for insular assemblages of birds and mammals facing global change

【岛屿鸟类和哺乳动物群落在全球变化下的差异脆弱性】

☐ Conservation Biology | Sat, 14 Mar 2026 04: [Latest] | 重要性: 36 分

☐ Abstract: Conservation Biology, EarlyView.

☐ 中文摘要:

本研究比较了面对全球变化（气候变化、入侵物种、栖息地丧失）时，岛屿鸟类和哺乳动物群落表现出的截然不同的脆弱性格局，揭示岛屿生物地理学在物种保护优先级确定中的差异性含义。

☐ 深度解读:

本研究发表于 Conservation Biology，通过系统比较全球岛屿生态系统中鸟类和哺乳动物对多重全球变化驱动因素的差异化响应，揭示了类群特异性保护需求。研究发现鸟类和哺乳动物在气候变化适应能力、入侵物种抵抗力等方面存在系统性差异，这对于岛屿保护区管理策略的制定具有重要指导意义。对中国南海岛屿的生物多样性保护管理和生态安全评估具有直接参考价值。

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70261

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181. Using resurvey data to predict changes in ecosystem functioning across protected and unprotected coastal dunes

☐ Conservation Biology | Sat, 14 Mar 2026 04: [Latest] | 重要性: 34 分

☐ Abstract: Conservation Biology, EarlyView.

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70256

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182. Advancing conservation breeding programs for marine invertebrates

☐ Conservation Biology | Wed, 11 Mar 2026 22: [Latest] | 重要性: 34 分

☐ Abstract: Conservation Biology, EarlyView.

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70253

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183. Electric stealth through bidirectional signal suppression in electric eels and their knifefish prey

☐ Current Biology [Latest] | 重要性: 34 分

☐ Abstract: Publication date: Available online 16 March 2026Source: Current BiologyAuthor(s): Lok Poon, William G.R. Crampton

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0960982226001776

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184. Guiding stakeholder negotiations in data-poor coastal planning with open-access spatial data

☐ Conservation Biology | Wed, 11 Mar 2026 22: [Latest] | 重要性: 32 分

☐ Abstract: Conservation Biology, EarlyView.

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70248

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185. Indigenous governance and the future of conservation

☐ Conservation Biology | Tue, 10 Mar 2026 23: [Latest] | 重要性: 32 分

☐ Abstract: Conservation Biology, EarlyView.

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70254

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186. Aerobic sister lineage of breviates has gene-rich mitochondrial genomes

☐ Current Biology [Latest] | 重要性: 32 分

☐ Abstract: Publication date: Available online 13 March 2026Source: Current BiologyAuthor(s): Anna Cho, John A. Burns, Tanja Woyke, Ramunas Stepanauskas, Nicole Poulton, Jeremy G. Wideman

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0960982226001752

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187. Sleep facilitates pattern separation through SK channel-mediated sparse coding

☐ Current Biology [Latest] | 重要性: 30 分

☐ Abstract: Publication date: Available online 16 March 2026Source: Current BiologyAuthor(s): Chien-Chun Chen, Yu-Chun Huang, Antonio Ortega, Raquel Suárez-Grimalt, Erik Tedre, El-Sayed Baz, Yifan Wu, Andrew C. Lin, Sha Liu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0960982226002058

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188. Receptor-like-kinase-interacting protein TOW stabilizes PIN transporters for auxin canalization

☐ Current Biology [Latest] | 重要性: 30 分

☐ Abstract: Publication date: Available online 13 March 2026Source: Current BiologyAuthor(s): Mingyue Li, Nikola Rydza, Ewa Mazur, Gergely Molnár, Tomasz Nodzyński, Jiří Friml

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0960982226001740

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189. Abrogation of presynaptic facilitation at hippocampal mossy fiber synapses disrupts neural ensemble activity and spatial memory

☐ Current Biology [Latest] | 重要性: 30 分

☐ Abstract: Publication date: Available online 12 March 2026Source: Current BiologyAuthor(s): Catherine Marneffe, Noelle Grosjean, Kyrian Nicolay-Kritter, Cécile Chatras, Evan R. Harrell, Ashley Kees, Christophe Mulle

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0960982226002034

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190. Assessing vertical stratification of bats and nocturnal insects to infer potential collision risk at wind turbines in central Thailand

☐ Biodiversity and Conservation | 2026-03-14 [Latest] | 重要性: 16 分

☐ Abstract: The global growth of wind energy can harm wildlife and ecosystems, especially airborne animals, which face high collision risks with turbine blades. While detailed knowledge of bird and bat collision risk exists for temperate regions, risk assessments for insects and bats in the Global South remain lacking. To fill this gap, we investigated the activity of bats and nocturnal insects in Thailand at different heights above the ground. To this end, we suspended ultrasonic detectors and sticky traps from a helium-filled balloon kite at 25, 50, 75, and 100 m above ground, and also mounted them on a 2 m post to record ground-level activity. We documented the rotor-swept zone of local wind turbines to range between 21 and 159 m above ground. Mostly bat species from the guild of open-space forager

☐ DOI: https://link.springer.com/article/10.1007/s10531-026-03307-x

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191. The role of an endangered vine parasite in promoting agritourism mediated conservation in Sabah, Malaysia

☐ Biodiversity and Conservation | 2026-03-13 [Latest] | 重要性: 16 分

☐ Abstract: Agritourism is increasingly viewed as an anchor for sustainable development in Sabah, Malaysia, extending beyond its well-known and established wildlife-based tourism industry. This commentary considers a future role of agritourism that, when combined with targeted conservation practices, has the potential to aid in the broader effort to protect Sabah's rich biodiversity and support local indigenous economies. While Rafflesia, which is frequently located on private land, is an iconic species that has long attracted international visitors, many regions in Sabah possess untapped potential for eco-focused agritourism within natural landscape settings. Agritourism that protects forest could also act as an impetus to engage in the re-wilding of areas via the re-introduction of declining and rar

☐ DOI: https://link.springer.com/article/10.1007/s10531-026-03315-x

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192. The Rpd3 histone deacetylase is a critical regulator of temperature-mediated morphogenesis and virulence in the human fungal pathogen Histoplasma

☐ PLOS Biology | 2026-03-17T14:00:00Z [Latest] | 重要性: 12 分

☐ Abstract: by Nebat Ali, Mark Voorhies, Rosa A. Rodriguez, Anita Sil Adaptive responses to environmental stimuli are integral to the survival and virulence of microbial pathogens. The thermally dimorphic human fungal pathogen Histoplasma senses temperature to transition between a mold form in soil and a pathogenic yeast in mammalian hosts. The contributions of chromatin-modifying enzymes to the ability of Histoplasma to appropriately respond to temperature have never been explored. Through chemical inhibition and genetics, we determined that the class I histone deacetylase (HDAC) RPD3 is required for normal Histoplasma yeast morphology at 37 °C. Rpd3 regulated the expression of key morphology-specific genes, including critical virulence factors and transcription factors (TFs), was required for normal

☐ DOI: https://journals.plos.org/plosbiology/article

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193. Moth, beetle, and bug assemblages are shaped differently across a tropical rainforest vertical gradient

☐ Biodiversity and Conservation | 2026-03-17 [Latest] | 重要性: 12 分

☐ Abstract: Tropical forests harbour a large proportion of Earth's insect biodiversity. Tropical forest structure produces layers of habitat from the cooler, wetter understorey to the hotter, drier canopy, which provide stratified resources for inhabitants. This gradient of biotic and abiotic factors results in vertically stratified insect communities, often with different assemblages in the understorey and canopy. Vertical distributions of insect taxa are shaped by their niche requirements, with narrower vertical ranges for species with more specialised life history strategies and abiotic requirements. Many studies of vertical stratification focus on single taxonomic orders, often overlooking less well-worked orders, such as Hemiptera, and comparisons of the responses of different taxa sampled at the

☐ DOI: https://link.springer.com/article/10.1007/s10531-026-03274-3

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194. Combining Population Genomics With Ancient DNA to Understand Island Colonisation History of the Madagascar Turtle Dove

☐ Molecular Ecology | Mon, 16 Mar 2026 23: [Latest] | 重要性: 9 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/mec.70299

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195. Population Genetic Structure and Colonisation History of the Widely Distributed Mangrove Avicennia marina Across the Western Indian Ocean

☐ Diversity and Distributions | Fri, 13 Mar 2026 22: [Latest] | 重要性: 8 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ddi.70147

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196. Next-generation agri-environment schemes: Integrating innovations in biodiversity monitoring, farming technologies and digital tools

☐ Biological Conservation [Latest] | 重要性: 8 分

☐ Abstract: Publication date: May 2026Source: Biological Conservation, Volume 317Author(s): Anna F. Cord, Frank Wätzold

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0006320726000698

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197. Factors Influencing Species Distribution Model Performance in Tropical Reef Fishes

☐ Diversity and Distributions | Wed, 11 Mar 2026 23: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/ddi.70160

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198. Decades of conservation and monitoring reveal population recovery in a globally important loggerhead rookery

☐ Biological Conservation [Latest] | 重要性: 6 分

☐ Abstract: Publication date: May 2026Source: Biological Conservation, Volume 317Author(s): Cassandra Roch, Carlos Angulo-Preckler, Kenydjeer Lima Rodrigues, Hugo Mann, Hector Barrios-Garrido, Natalie Wildermann, Luis F. López-Jurado, Ana Liria-Loza, Maria Medina-Suarez, Carlos M. Duarte

☐ DOI: https://www.sciencedirect.com/science/article/pii/S000632072600090X

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199. NATURA 2000: Synergies and trade-offs between process-oriented conservation and the protection of threatened species

☐ Biological Conservation [Latest] | 重要性: 6 分

☐ Abstract: Publication date: May 2026Source: Biological Conservation, Volume 317Author(s): Erich Tasser, Ingrid Loacker, Chiara Paniccia, Jonas Sommer, Elena Tello García, Brenda Maria Zoderer

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0006320726000935

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200. The World Could Reach 30% Protection by 2030, and Still Fail to Conserve Biodiversity without Effective and Well-located Protected Areas

☐ Conservation Letters | Mon, 09 Mar 2026 23: [Latest] | 重要性: 6 分

☐ Abstract: Conservation Letters, Volume 19, Issue 2, March/April 2026.

☐ DOI: https://conbio.onlinelibrary.wiley.com/doi/10.1111/con4.70030

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201. Drivers of Viral Diversity and Sharing in Marine Mammals

☐ Molecular Ecology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/mec.70294

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202. Does Allometric Scaling Improve Estimates of Population Abundance Based on Environmental DNA?

☐ Molecular Ecology | Mon, 16 Mar 2026 03: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/mec.70303

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203. Systematic identification of Salmonella T6SS effectors uncovers diverse new families and lipid-targeting activities

☐ PLOS Biology | 2026-03-17T14:00:00Z [Latest] | 重要性: 6 分

☐ Abstract: by Gianlucca G. Nicastro, Stephanie Sibinelli-Sousa, Julia T. Hespanhol, Thomas W. C. Santos, Joseph P. Munoz, Rosangela S. Santos, Blanca M. Perez-Sepulveda, Sayuri Miyamoto, L. Aravind, Robson F. de Souza, Ethel Bayer-Santos Bacterial warfare is a widespread phenomenon in which bacteria deploy toxins to inhibit or kill competitors. These toxins disrupt essential cellular processes, and their diversification is driven by an evolutionary arms race involving toxin and immunity gene acquisition. Here, we used in-silico approaches to analyze genomes from the 10k Salmonella Project and identify effectors secreted via the Type VI Secretion System (T6SS). We uncovered 128 candidates distributed across diverse Salmonella serovars and other bacterial species. Among them, Tox-Act1 was selected for

☐ DOI: https://journals.plos.org/plosbiology/article

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204. Comparative embryogenesis of two salp species reveals rogue development and evolutionary divergence from sessile tunicates

☐ PLOS Biology | 2026-03-17T14:00:00Z [Latest] | 重要性: 6 分

☐ Abstract: by Marie Lebel, Alexandre Alié, Patrick Lemaire, Stefano Tiozzo Tunicates are the closest living relatives of vertebrates. Recent phylogenies place the little-studied, free-swimming thaliaceans—including salps—within sessile ascidians, highlighting a remarkable ecological transition. Historical reports hinted at a parallel developmental shift. Salp embryogenesis diverges from that of ascidians and involves unique maternal cells called calymmocytes. Here, we provide foundational resources for two distantly related salp species, Salpa fusiformis and Thalia democratica. Using advanced microscopy, we generated developmental staging tables showing that while embryogenesis is stereotyped within species, it differs in cleavage patterns and blastomere positioning between them. We traced the origin

☐ DOI: https://journals.plos.org/plosbiology/article

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205. Is urban part-time lighting a mitigation measure for a common amphibian? A case study on Bufo spinosus

☐ Biodiversity and Conservation | 2026-03-12 [Latest] | 重要性: 6 分

☐ Abstract: Artificial light at night is an expanding phenomenon that affects biodiversity at the global scale. Mitigation measures consisting in switching off public lights during a part of the night are increasingly used. Whether such measures actually benefit biodiversity, beyond reducing the energy budget of municipalities, remains largely untested. We investigated the effect of part-time lighting on the behaviour of adult spiny toads (Bufo spinosus). We analysed the distance travelled and the use of shelters by individuals in arenas kept in the dark or exposed to light all night or part of it, i.e. excluding the 23:00−05:00 time period. Treatment had a relatively small effect on distance travelled during the night but shelters were increasingly used as the duration of the lit period at night incr

☐ DOI: https://link.springer.com/article/10.1007/s10531-026-03313-z

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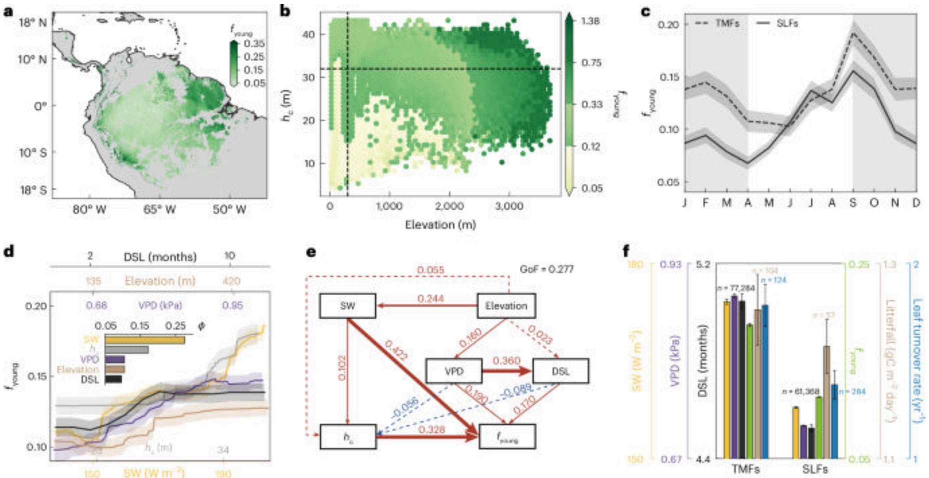
【植物与农业】

206. Amazon rainforests are rejuvenating their canopies by producing more photosynthetically efficient young leaves under climate change

【气候变化下亚马逊雨林通过大量产生高光合效率嫩叶更新林冠】

☐ Nature Plants | 2026-03-09 [Latest] | 重要性: 51 分

☐ Graphical Abstract:



☐ Abstract: Nature Plants, Published online: 09 March 2026; doi:10.1038/s41477-026-02240-9The authors mapped the continental-scale fraction of age-dependent leaf area index and revealed a widespread increase in the fraction of photosynthetically efficient young leaves across the majority of Amazon rainforests over the past decades.

☐ 中文摘要:

研究者绘制了大陆尺度上年龄依赖叶面积指数的分布图，揭示了亚马逊大部分雨林中光合高效嫩叶比例的广泛增加。这一发现表明热带雨林正在通过加速叶片周转来响应气候变化，并揭示了雨林维持光合能力的适应机制。

☐ 深度解读:

本文发表于 Nature Plants，基于卫星遥感和地面观测，揭示了亚马逊雨林应对气候变化的一种重要适应策略——加速林冠叶片周转以维持整体光合效能。这一发现挑战了此前关于热带雨林在气候变化下净初级生产力必然下降的悲观预测，表明雨林系统具有一定的自我调节能力。然而研究同时指出这种适应是有代价的：更高的叶片周转速率意味着更大的氮磷营养消耗，长期可能导致养分限制，研究结果对全球碳循环预测具有重大修正意义。

☐ DOI: https://doi.org/10.1038/s41477-026-02240-9

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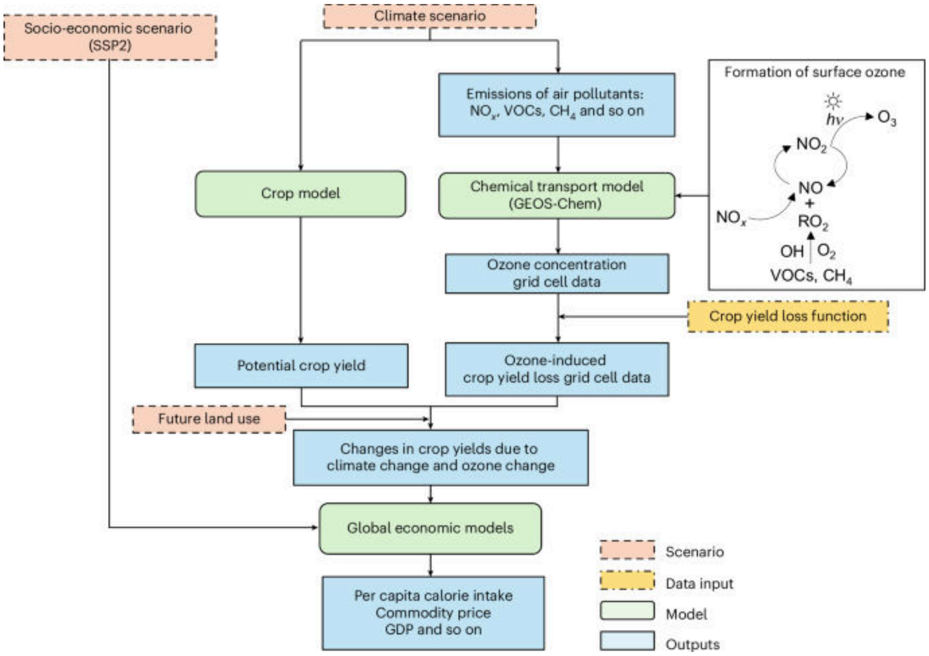
207. Ozone pollution reduction partially offsets the negative impact of climate change mitigation efforts on global hunger

【臭氧污染减少部分抵消气候减缓措施对全球饥饿的负面影响】

☐ Nature Food | 2026-03-16 [Latest] | 重要性: 51 分

☐ Graphical Abstract:



☐ Abstract: Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01322-3A multi-model intercomparison using six global agro-economic models explores how ozone reductions arising from climate change mitigation could impact crop yields. Yield changes are projected to 2050, linking reduced ozone to food security and the alleviation of hunger.

☐ 中文摘要:

通过六个全球农业经济模型的多模型比较研究，探讨气候变化减缓行动带来的臭氧浓度降低如何影响农作物产量。研究预测到 2050 年的产量变化，将臭氧减少与农业生产力和粮食安全相关联，揭示气候政策的粮食安全协同效益。

☐ 深度解读:

本研究发表于 Nature Food，采用多模型集成方法，首次系统量化了气候减缓政策通过降低地表臭氧浓度对全球农业产量和粮食安全的间接协同效益。研究发现减少化石燃料燃烧不仅可直接降低温室气体排放，还通过减少臭氧前体物排放改善作物生长环境，预计可使全球主要粮食作物产量提升 1-5%。这一双重效益对中国等农业大国尤为重要，为推进碳中和政策提供了额外的粮食安全协同论据。

☐ DOI: https://doi.org/10.1038/s43016-026-01322-3

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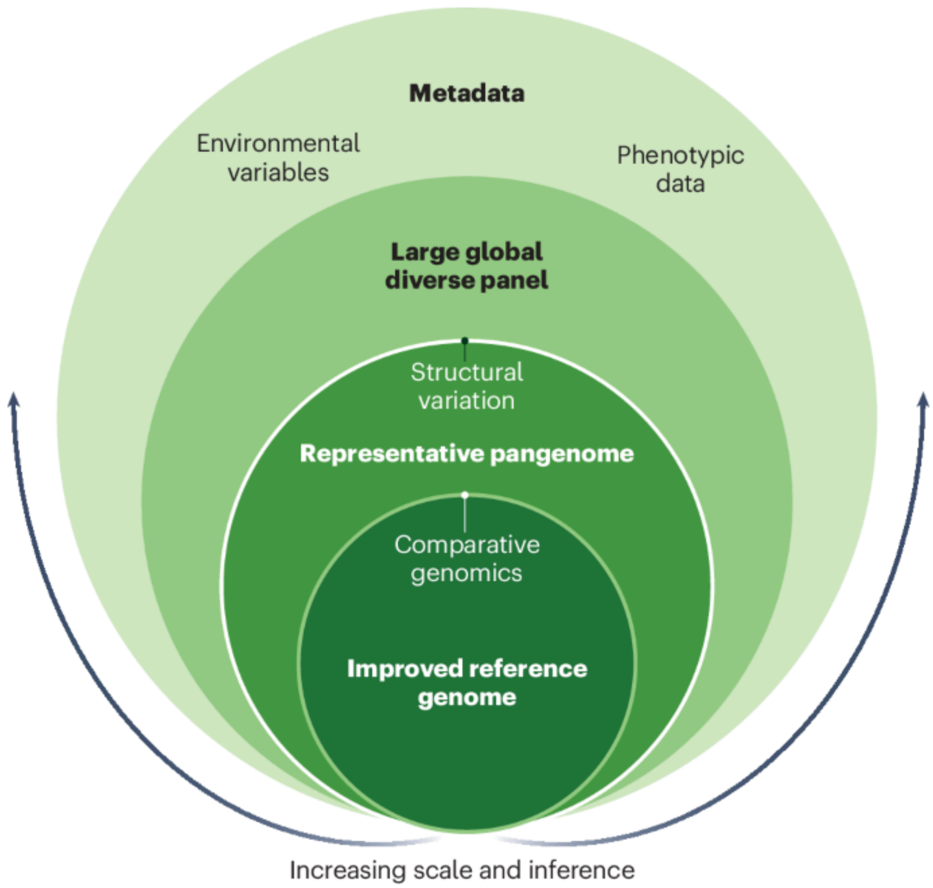
208. Pangenomics for sorghum improvement

☐ Nature Plants | 2026-03-13 [Latest] | 重要性: 43 分

☐ Graphical Abstract:



☐ Abstract: Nature Plants, Published online: 13 March 2026; doi:10.1038/s41477-026-02266-zA multilayered pangenomic resource for Sorghum bicolor, integrating an improved reference genome, a representative 33-accession pangenome and a global panel of nearly 2,000 lines, establishes a powerful platform for trait discovery and genetic improvement.

☐ DOI: https://doi.org/10.1038/s41477-026-02266-z

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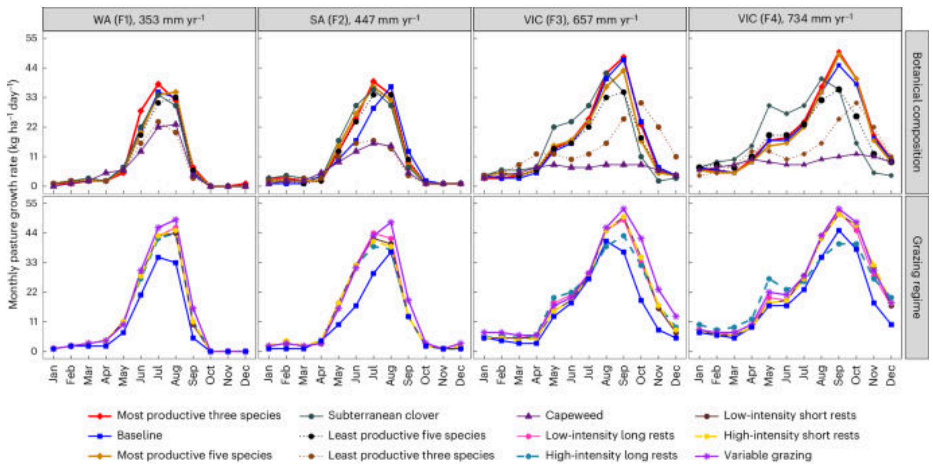
209. Regenerative agriculture improves productivity and profitability while reducing greenhouse gas emissions on Australian sheep farms

【再生农业在澳大利亚绵羊农场中提高生产力和盈利能力同时减少温室气体排放】

☐ Nature Food | 2026-03-13 [Latest] | 重要性: 40 分

☐ Graphical Abstract:



☐ Abstract: Nature Food, Published online: 13 March 2026; doi:10.1038/s43016-026-01331-2Regenerative agriculture promises lower emissions and healthier soils, but results vary by environment and practice. Across four diverse rainfall zones, this study finds that practices that yielded the largest soil carbon gains and greatest greenhouse gas mitigation were not always the most profitable, underscoring the need for context-specific evaluation using multiple sustainability metrics.

☐ 中文摘要:

再生农业承诺降低排放和改善土壤健康，但结果因环境和实践而异。在四个多样化降雨区的研究中发现，产生最大土壤碳收益的实践同时也带来了温室气体减排效益，表明在澳大利亚牧羊农场实施再生农业可以在不牺牲生产力的前提下实现气候目标。

☐ 深度解读:

本研究发表于 Nature Food，基于澳大利亚四个典型降雨区的跨站点比较研究，系统评估了再生农业实践的环境和经济双重效益。研究关键发现是土壤固碳量最大的管理实践同时也是温室气体减排效果最显著的，打破了减排必然降低产量和利润的固有认知。这一协同效益发现对中国农业减排政策具有重要参考价值，特别是对正在推进的农业绿色发展和农业碳汇项目，同时为全球性气候智慧农业实践提供了有力证据。

☐ DOI: https://doi.org/10.1038/s43016-026-01331-2

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210. Plant Community Responses to Long-Term Nutrient Additions Interact With Elevation and Vegetation Type in a Subarctic Tundra

☐ Journal of Vegetation Science | Wed, 04 Mar 2026 21: [Latest] | 重要性: 38 分

☐ Abstract: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70125

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211. Effects of Ungulate Herbivores on Temperate Forest Understory Vegetation—Implications From a Large-Scale Wildlife Exclosure Experiment in Central Europe

☐ Journal of Vegetation Science | Sun, 08 Mar 2026 21: [Latest] | 重要性: 36 分

☐ Abstract: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70128

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212. Grass inflorescence morphodynamics guides yield improvement in wheat

☐ Nature Plants | 2026-03-11 [Latest] | 重要性: 34 分

☐ Abstract: Nature Plants, Published online: 11 March 2026; doi:10.1038/s41477-026-02246-3Morphodynamic modelling reveals how timing and fate shape diverse grass inflorescences, including wheat supernumerary spikelets. Using this model, the authors identify an early-heading paired-spikelet mutant, which exhibits robust yield gains.

☐ DOI: https://doi.org/10.1038/s41477-026-02246-3

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213. A systematic review of sustainable food systems identifies socio-economic pathways driving food systems transformations

☐ Nature Food | 2026-03-16 [Latest] | 重要性: 34 分

☐ Abstract: Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01317-0Studies on food systems transformations are heterogeneous. This systematic review develops pathways aimed at six key food systems actors across socio-economic contexts that can shift food systems towards greater sustainability.

☐ DOI: https://doi.org/10.1038/s43016-026-01317-0

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214. Differentiation of the Seed Regeneration Niche Along a Small-Scale Plant Zonation in Mediterranean Temporary Ponds

☐ Journal of Vegetation Science | Tue, 03 Mar 2026 05: [Latest] | 重要性: 34 分

☐ Abstract: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70122

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215. Neighbourly Dispute at the Edge of Life: Species Interactions Among Antarctic Mosses

☐ Journal of Vegetation Science | Fri, 27 Feb 2026 06: [Latest] | 重要性: 34 分

☐ Abstract: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70123

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216. Cross-kingdom RNA interference promotes arbuscular mycorrhiza development

☐ Nature Plants | 2026-03-11 [Latest] | 重要性: 32 分

☐ Abstract: Nature Plants, Published online: 11 March 2026; doi:10.1038/s41477-026-02247-2The authors provide evidence that symbiotic fungi forming widespread arbuscular mycorrhiza symbioses use cross-kingdom RNA interference to silence plant genes and promote their colonization of host roots.

☐ DOI: https://doi.org/10.1038/s41477-026-02247-2

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217. Resurrecting the American chestnut

☐ Nature Plants | 2026-03-09 [Latest] | 重要性: 32 分

☐ Abstract: Nature Plants, Published online: 09 March 2026; doi:10.1038/s41477-026-02261-4Resurrecting the American chestnut

☐ DOI: https://doi.org/10.1038/s41477-026-02261-4

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218. The socioeconomic conditions that drive transformation

☐ Nature Food | 2026-03-16 [Latest] | 重要性: 32 分

☐ Abstract: Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01320-5Food systems encompass broad socioeconomic contexts involving several actors and sectors of society. Now, seven types of transition and six socioeconomic drivers have been identified that can facilitate or hinder sustainable food system transformations.

☐ DOI: https://doi.org/10.1038/s43016-026-01320-5

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219. Rethinking sugar

☐ Nature Food | 2026-03-13 [Latest] | 重要性: 32 分

☐ Abstract: Nature Food, Published online: 13 March 2026; doi:10.1038/s43016-026-01326-zSugar is often portrayed in public discourse as a dietary villain to be avoided at all costs. This framing sits uneasily with biological evolution, culture and history.

☐ DOI: https://doi.org/10.1038/s43016-026-01326-z

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220. The Apple Does Not Fall Far From the Tree and How This Affects the Abiotic Stress Tolerance in Juvenile Trees

☐ Journal of Vegetation Science | Sat, 07 Mar 2026 21: [Latest] | 重要性: 32 分

☐ Abstract: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70126

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221. Effects of Irrigation Water Sources on Soil Fertility, Heavy Metal Accumulation in both Soil and Rice (Oryza sativa L.)

☐ Plant and Soil | 2026-03-15 [Latest] | 重要性: 27 分

☐ Abstract: Aims This study assessed the comparative impact of irrigation with sewage wastewater (SWW) and mixed sewage-industrial wastewater (SIWW) on soil fertility indicators and heavy metal accumulation in clayey soils and rice grains, using Nile water (NW) irrigation as a control. Materials and Methods Samples were collected from 21 sites in Dakahlia Governorate, Egypt, representing three irrigation sources: seven sites each irrigated by SWW, SIWW, and NW. Results Soils and rice grains from wastewater-irrigated sites showed significantly elevated levels of salinity, nutrients, organic matter, and bioavailable heavy metals compared to the Nile water control. Soil pH was the sole exception, showing no significant difference. SWW sources exhibited the highest levels of salinity, biochemical oxygen d

☐ DOI: https://link.springer.com/article/10.1007/s11104-026-08383-0

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222. Fulvic acids as biostimulants in plant – soil systems: mechanistic insights into nutrient chelation, transport, and stress tolerance

☐ Plant and Soil | 2026-03-16 [Latest] | 重要性: 23 分

☐ Abstract: Modern agriculture must enhance crop productivity while reducing environmental degradation associated with inefficient fertilizer use and increasing abiotic stress. Fulvic acid (FA), a low-molecular-weight humic fraction enriched in oxygen-containing functional groups, has emerged as a promising biostimulant for improving nutrient-use efficiency and stress resilience. Owing to its high solubility, pH-responsive ionization, and metal-chelating capacity, FA regulates nutrient speciation and mobility, enhancing the availability of iron (Fe), zinc (Zn), potassium (K), and phosphorus (P) across diverse soil environments. At the plant level, FA promotes nutrient acquisition primarily through bioenergetic and signaling modulation. Evidence consistently supports stimulation of plasma membrane H^+-A

☐ DOI: https://link.springer.com/article/10.1007/s11104-026-08461-3

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223. Co-designing a research agenda for UK agroforestry using a multi-actor approach

☐ Agronomy for Sustainable Development | 2026-02-27 [Latest] | 重要性: 19 分

☐ Abstract: There is growing recognition of agroforestry's potential to help mitigate and provide resilience to the climate and biodiversity crises. Beyond its environmental benefits, agroforestry can also enhance production and profits, making it a sustainable farming solution that is scalable. Despite this, uptake within Europe is low, and many knowledge gaps remain that need to be addressed to promote adoption and optimize the management and implementation of agroforestry systems. We co-developed a research agenda for agroforestry using a multi-actor approach and a modified Delphi method in 2023. 156 UK-based stakeholders contributed to this process, including farmers, advisors, policy makers, NGOs, and researchers. An initial list of 238 research priorities (high-priority research questions) was s

☐ DOI: https://link.springer.com/article/10.1007/s13593-026-01089-8

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224. Nitrogen fertilization surpasses rapeseed species in shaping rhizosphere enzyme activities and bacterial community structure

☐ Plant and Soil | 2026-03-17 [Latest] | 重要性: 19 分

☐ Abstract: Background and aims Rapeseed, a globally important oil crop with high nitrogen demand, exhibits rhizosphere microbial responses influenced by both nitrogen fertilization and species. However, the differential effects of nitrogen and rapeseed species on root morphology, rhizosphere enzyme activity, and microbial communities have not been adequately evaluated. Methods A rhizo-box experiment was conducted over 21 days using clay loam soil from a long-term paddy field. We compared the effects of urea application (0 vs. 0.12 g N kg^{-1} dry soil) and two rapeseed species (Brassica napus, cabbage-type; Brassica juncea, mustard-type) on root morphology, rhizosphere enzyme activity, and bacterial community structure. By integrating in situ soil zymography and bacterial amplicon sequencing, we compreh

☐ DOI: https://link.springer.com/article/10.1007/s11104-026-08448-0

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225. Optimizing straw return management to suppress plant disease and promote plant growth. A meta-analysis

☐ Agronomy for Sustainable Development | 2026-03-03 [Latest] | 重要性: 18 分

☐ Abstract: Straw return improves soil and nutrient cycling but may aggravate diseases if mismanaged. However, the comprehensive impact of different straw return methods on plant diseases remains unclear. This study aims to clarify the effects of straw return on plant diseases and explore appropriate straw return methods. This study evaluated the effect of straw return on the reduction of plant diseases by summarizing 1352 pairs of observations from 165 peer-reviewed publications. It further analyzed the effects of straw return conditions, disease types, and plant species on plant diseases. Overall, the study's findings showed that, with significant variation among return treatments, straw return reduced plant disease severity by 40.55% and increased plant biomass by 29.05%. Direct straw return of dif

☐ DOI: https://link.springer.com/article/10.1007/s13593-026-01094-x

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226. Positive shrub-soil feedbacks enhance growth of alpine juniper shrubs in the central Himalayas

☐ Plant and Soil | 2026-03-16 [Latest] | 重要性: 18 分

☐ Abstract: Background and aims Plant-soil feedbacks impact climate-driven plant expansions. However, how canopy-induced soil heterogeneity affects these feedbacks, particularly in climatically stressful high-elevation sites, remains unclear. Methods This study investigates whether soil carbon (C), soil nitrogen (N), carbon-to-nitrogen (C/N) ratios, soil pH, and soil water holding capacity influence alpine juniper radial growth in 22 sites across the central Himalayas by comparing two microsites (under canopy and bare ground). Results Soil C, N ,C/N, and water holding capacity were higher under the canopy than outside canopy, while soil pH exhibited no significant differences between the two situations. The growth rate of juniper shrubs was negatively correlated with the soil C/N ratio measured under

☐ DOI: https://link.springer.com/article/10.1007/s11104-026-08464-0

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227. FarmSTEPS, a model for spatial-temporal exploration of diversified and sustainable crop sequence plans for crop farms

☐ Agronomy for Sustainable Development | 2026-03-17 [Latest] | 重要性: 16 分

☐ Abstract: Crop diversification could enhance economic and environmental sustainability of crop farming, but achieving this requires careful planning that accounts for crop interrelationships in a rotation or spatial layout. Several models have been devised to design diversified crop rotations or crop spatial layouts, but they often fail to capture farm settings, production constraints, and sustainability objectives specific to a farm at both temporal and spatial scales. To fill this gap, we developed a novel rule-based whole-farm planning model, FarmSTEPS, for systematically exploring diversified and sustainable farm crop sequence plans in a spatial-temporally explicit manner. FarmSTEPS comprises multiple farm characterization modules, an algorithm for generating farm crop sequence plans, and a mult

☐ DOI: https://link.springer.com/article/10.1007/s13593-026-01097-8

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228. Impacts of long-term cassava-based conservation agriculture systems on soil greenhouse gas emissions in Cambodia

☐ Agriculture Ecosystems & Environment [Latest] | 重要性: 15 分

☐ Abstract: Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Vira Leng, Laurent Thuriès, Vang Seng, Florent Tivet, Phearum Mark, Chhay Ngin, Try Yorn, Titouan Filloux, Pascal Lienhard, Johan Six, Lyda Hok, Stéphane Boulakia, Manuel Reyes, P.V. Vara Prasad, Rémi Cardinael

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167880926001507

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229. Phylogenetic relatedness of a focal plant to weed species and field crop species affects its seed production through insect-mediated trophic interactions

☐ Agriculture Ecosystems & Environment [Latest] | 重要性: 14 分

☐ Abstract: Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Nadia K. Toukem, Emeline Felten, Emilien Laurent, Diana Bottini, Samuel Pouget, Stephane Cordeau, Sandrine Petit, Adam J. Vanbergen

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167880926001349

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230. Precision planting of cereal rye cover crop improves sweet corn yield and farm benefits

☐ Agronomy for Sustainable Development | 2026-03-17 [Latest] | 重要性: 14 分

☐ Abstract: Including cereal rye (Secale cereale L.) in sweet corn (Zea mays L.) production systems provides environmental and soil benefits. However, cereal rye often decreases sweet corn yield and thus farm profit. A novel approach to reduce this negative effect is precision planting of cereal rye, which skips the sweet corn row to minimize crop interactions, potentially increasing nitrogen (N) availability, and reducing negative effects of cereal rye on sweet corn. A four-site-yr trial was conducted during the 2022-2024 growing seasons, the first known evaluation of precision-planted cereal rye compared to a no-cover crop control and solid-planted cereal rye. The study evaluated (i) cereal rye morphology, biomass production and its quality, (ii) sweet corn plant population, morphology, and marketab

☐ DOI: https://link.springer.com/article/10.1007/s13593-026-01095-w

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231. Mixed shrub forests are superior in enriching desert vegetation diversity: evidence for an indirect pathway mediated by soil microorganisms

☐ Plant and Soil | 2026-03-16 [Latest] | 重要性: 12 分

☐ Abstract: Aims Shrub expansion is a major strategy for vegetation restoration in desertified regions and has been shown to enhance understorey herbaceous diversity. However, the role of soil microorganisms in this process remains unclear. Investigating the role of soil microorganisms in shrub expansion-driven changes in plant diversity will therefore improve our understanding of the underlying mechanisms. Methods This study focused on three communities in the Ulan Buh Desert with differing degrees of shrub expansion (mixed shrub forest, pure shrub forest, and pure herbaceous communities) to investigate the effects of shrub expansion on plant and microbial communities and to quantify the role of soil microbial communities in mediating the impact of shrub expansion on plant diversity. Results The soil

☐ DOI: https://link.springer.com/article/10.1007/s11104-026-08401-1

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232. Assessing the interkingdom response of soil prokaryotes, fungi and protists to prairie restoration on retired agricultural lands

☐ Agriculture Ecosystems & Environment [Latest] | 重要性: 10 分

☐ Abstract: Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Micaela Tosi, Kevin MacColl, Dasiel Obregon, Andrew S. MacDougall, Hafiz Maherali, Kari E. Dunfield

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167880926001544

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233. Integrative Genomic Analysis Reveals Modular Control of Mycorrhizal Fungi and Rhizobia Symbiosis in Soybean

☐ Plant Cell & Environment | Thu, 12 Mar 2026 20: [Latest] | 重要性: 9 分

☐ Abstract: Plant, Cell &Environment, EarlyView.

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/pce.70491

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234. A conceptual framework for simultaneous optimization of integrated climate scenario data and phenology models

☐ Agronomy for Sustainable Development | 2026-03-12 [Latest] | 重要性: 8 分

☐ Abstract: Climate-resilient crop varieties are exceedingly important for global food security. High-accuracy predictions of crop phenology in future climate scenarios require a more precise description of putative crop-environment interactions. While the accuracy of these predictions is affected by three primary sources of uncertainty —the phenology model, the climate scenario data, and their interaction —commonly used approaches consider only the phenology model as a source of uncertainty, while uncertainties arising from climate scenario data and their interaction with the model are commonly neglected. To address this gap, we developed a novel dynamic crop phenology model framework that simultaneously integrates uncertainties arising from all three sources. This model framework was developed usi

☐ DOI: https://link.springer.com/article/10.1007/s13593-026-01091-0

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235. Crop rotation and nitrogen management achieve a trade-off among crop productivity, net income and soil CO2 emission in the Hexi Oasis irrigation area

☐ Field Crops Research [Latest] | 重要性: 6 分

☐ Abstract: Publication date: 15 May 2026Source: Field Crops Research, Volume 342Author(s): Lili Yang, Xiaoyuan Bao, Hong Fan, Congcong Guo, Fuyang Cui, Jie Zhou, Chengxin Bai, Qiang Chai, Cai Zhao

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0378429026001103

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236. Wheat breeding during and after the “green revolution”contributed to the reduced use of elite nitrogen metabolism alleles linked to nitrogen use efficiency

☐ Plant Journal | Mon, 16 Mar 2026 18: [Latest] | 重要性: 6 分

☐ DOI: https://onlinelibrary.wiley.com/doi/10.1111/tpj.70792

【土壤与生物地球化学】

237. Alternative dynamic regimes of marine biogeochemical models in perturbed environments

【扰动环境中海洋生物地球化学模型的替代动态状态】

☐ Biogeosciences | Tue, 17 Mar 2026 08: [Latest] | 重要性: 59 分

☐ Abstract: Alternative dynamic regimes of marine biogeochemical models in perturbed environments Guido Occhipinti, Davide Valenti, and Paolo Lazzari Biogeosciences, 23, 2003–2021, https://doi.org/10.5194/bg-23-2003-2026, 2026 Due to climate change shifts in ecosystem structure and function have been increasingly documented in marine ecosystems around the globe. We tested whether a marine biogeochemical model can predict shifts to alternative regimes in plankton and biogeochemical processes under environmental perturbations. Simulations show that perturbations can drive the system into new regimes, with responses that are either reversible or hysteretic, depending on the type and intensity of the disturbance.

☐ 中文摘要:

由于气候变化，生态系统结构和功能的转变正在加速。本研究分析了扰动环境中海洋生物地球化学模型可能出现的替代稳定状态，揭示海洋营养级联响应气候变化可能存在非线性突变，为理解海洋生态系统的临界点提供了理论框架。

☐ 深度解读:

本研究发表于 Biogeosciences，通过数学分析和模型模拟，揭示海洋生物地球化学系统在气候变化压力下可能存在多个替代稳定状态，即生态系统可能发生不可逆的状态跳跃。研究识别了触发这种状态转换的关键阈值（温度、酸化、富营养化），对预测未来海洋生产力变化和碳泵效率具有重要意义。这一理论框架为理解海洋生态系统的临界点风险提供了新视角，对中国黄海、东海等近海生态系统的长期预测和管理决策具有参考价值。

☐ DOI: https://doi.org/10.5194/bg-23-2003-2026

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238. Photochemistry of the sea-surface microlayer (SML) influenced by a phytoplankton bloom: a mesocosm study

☐ Biogeosciences | Mon, 16 Mar 2026 08: [Latest] | 重要性: 40 分

☐ Abstract: Photochemistry of the sea-surface microlayer (SML) influenced by a phytoplankton bloom: a mesocosm study Olenka Jibaja Valderrama, Daniele Scheres Firak, Thomas Schaefer, Manuela van Pinxteren, Khanneh Wadinga Fomba, and Hartmut Herrmann Biogeosciences, 23, 1965–1985, https://doi.org/10.5194/bg-23-1965-2026, 2026 The present study explores the influence of biological activity in the photochemistry of the sea-surface microlayer (SML) and its implications for the emission of volatile organic compounds (VOCs) to the marine atmosphere. Experimental evidence of enhanced photochemical activity of carbonyl compounds in the SML is provided, particularly in periods of higher biological productivity, thereby offering new insights to integrate biological processes and photochemistry in the air-sea bo

☐ DOI: https://doi.org/10.5194/bg-23-1965-2026

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239. Litter vs. Lens: Evaluating LAI from Litter Traps and Hemispherical Photos Across View Zenith Angles and Leaf Fall Phases

☐ Biogeosciences | Thu, 12 Mar 2026 08: [Latest] | 重要性: 40 分

☐ Abstract: Litter vs. Lens: Evaluating LAI from Litter Traps and Hemispherical Photos Across View Zenith Angles and Leaf Fall Phases Simon Lotz, Teja Kattenborn, Julian Frey, Salim Soltani, Anna Göritz, Tom Jaksztat, and Negin Katal Biogeosciences, 23, 1949–1963, https://doi.org/10.5194/bg-23-1949-2026, 2026 Digital hemispherical photography(DHP) is a valuable tool for monitoring leaf area index(LAI), a key factor in ecosystem productivity and climate interactions. We compared DHP with litter traps in a temperate forest and found that at a view zenith angle around 20°, both methods aligned best. We applied a calibration model to assess site variability which significantly improved accuracy. Our findings enhance the reliability of ground-based LAI monitoring, supporting better ecosystem assessments.

☐ DOI: https://doi.org/10.5194/bg-23-1949-2026

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240. Spatial heterogeneity of GHG dynamics across an estuarine ecosystem

☐ Biogeosciences | Thu, 12 Mar 2026 08: [Latest] | 重要性: 38 分

☐ Abstract: Spatial heterogeneity of GHG dynamics across an estuarine ecosystem Nicolas-Xavier Geilfus, Bruno Delille, Anna Villnäs, and Alf Norkko Biogeosciences, 23, 1931–1948, https://doi.org/10.5194/bg-23-1931-2026, 2026 Coastal ecosystems strongly influence the global carbon cycle but remain poorly quantified. We measured surface pCO_2, CH_4, and N_2O concentrations in southwest Finland across an estuarine ecosystem. Greenhouse gases concentrations varied with salinity and habitat type. Riverine inputs and mixing with seawater, and primary production shaped greenhouse gases dynamics, emphasizing benthic control as a key yet uncertain driver of coastal carbon fluxes.

☐ DOI: https://doi.org/10.5194/bg-23-1931-2026

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241. The Farmer and Soil Health: Understanding the Foundation for Sustainable Soil Management

☐ European Journal of Soil Science | Sun, 15 Mar 2026 21: [Latest] | 重要性: 34 分

☐ Abstract: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70294

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242. The Role of Autonomous Mechanical Weeding Robots in Climate-Smart Soil Management: A Scoping Review

☐ European Journal of Soil Science | Wed, 11 Mar 2026 04: [Latest] | 重要性: 34 分

☐ Abstract: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70302

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243. Trade-Off Analysis Between Environmental Effects and Profitability in Agriculture: A Finnish Case-Study

☐ European Journal of Soil Science | Wed, 11 Mar 2026 00: [Latest] | 重要性: 34 分

☐ Abstract: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70300

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244. Soil Management Effects on Grapevine Water Uptake Depth in a Mediterranean Vineyard

☐ European Journal of Soil Science | Mon, 09 Mar 2026 00: [Latest] | 重要性: 34 分

☐ Abstract: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70301

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245. The distribution and isotopomeric characterization of nitrous oxide in the Eastern Gotland Basin (central Baltic Sea)

☐ Biogeosciences | Tue, 17 Mar 2026 08: [Latest] | 重要性: 32 分

☐ Abstract: The distribution and isotopomeric characterization of nitrous oxide in the Eastern Gotland Basin (central Baltic Sea) Pratirupa Bardhan, Claudia Frey, Gregor Rehder, and Hermann W. Bange Biogeosciences, 23, 1987–2002, https://doi.org/10.5194/bg-23-1987-2026, 2026 Nitrous oxide (N2O), a potent greenhouse gas, is released from coastal seas & estuaries, yet we don't fully understand how it is formed and consumed. In this study we collected water from several sites in the central Baltic Sea. N2O came from ammonia in oxic waters. Deep waters with low to no oxygen noted more active N2O cycling. The seafloor was a source in some areas. Typically N2O is produced by bacteria, but our results indicate possibility of other players like fungi or chemical reactions.

☐ DOI: https://doi.org/10.5194/bg-23-1987-2026

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246. Evaluating Soil Aeration Dynamics in a Non-Rigid Vertisol Through Shrink and Swelling Curves During Wetting-Drying Cycle

☐ European Journal of Soil Science | Sun, 15 Mar 2026 16: [Latest] | 重要性: 32 分

☐ Abstract: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70304

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247. Nitrate sources and transformation pathways in salt-affected agricultural ditches: biogeochemical responses along contrasting salinity gradients

☐ Biogeochemistry | 2026-03-09 [Latest] | 重要性: 18 分

☐ Abstract: Agricultural nitrate pollution, largely associated with high-intensity fertilizer application and untreated or insufficiently treated waste inputs, has become a major driver of water quality degradation worldwide. Agricultural ditches, as key recipients of surface runoff and lateral seepage, function as both transport conduits and biogeochemical hotspots for nitrogen transformation. In salt-affected ditches, high nitrogen inputs and salinity can interact to enhance nitrate accumulation and mobility, thereby increasing the potential for downstream transport. To bridge this knowledge gap, we investigated two representative saline agricultural ditches located in northwestern China, a region characterized by an arid climate and saline–alkaline soils. We employed stable isotope analysis (δ15N-N

☐ DOI: https://link.springer.com/article/10.1007/s10533-026-01318-y

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248. Mobilized dissolved organic matter provides niche space for prokaryotes in the deep subterranean estuary of a sandy beach

☐ Biogeochemistry | 2026-03-17 [Latest] | 重要性: 16 分

☐ Abstract: A fraction of organic carbon in beach sands is mobilizable as dissolved organic matter (DOM). The molecular composition of this mobilizable DOM and its relation to microbial metabolism is central for biogeochemical processes in beach ecosystems, yet, still poorly understood. To identify these DOM-microbe interactions, we analyzed two continuous 24-m-long sediment cores from the high-energy beach of Spiekeroog Island, Germany. The beach is characterized by a deep subterranean estuary (STE), where sediments are flushed by fresh, saline and brackish water. Even though beach sands are generally low in organic carbon, we found that concentrations of mobilizable DOM from sediments were approximately 16 times higher than the in-situ groundwater concentrations. Ultrahigh-resolution mass spectromet

☐ DOI: https://link.springer.com/article/10.1007/s10533-026-01315-1

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249. No indications for a priming effect on soil organic carbon mineralization in a temperate river system

☐ Biogeochemistry | 2026-03-07 [Latest] | 重要性: 16 分

☐ Abstract: Globally, rivers are vital conduits transporting and processing terrestrial carbon, and are generally considered to act as source of carbon dioxide (CO_2) towards the atmosphere. A large amount of soil organic carbon (SOC) is transferred from the land surface to river systems each year, where it mixes and interacts with the autochthonous carbon pool (i.e., produced in-stream via photosynthesis). The latter has been suggested to be more labile and to potentially affect—positively or negatively—the mineralization rate of the more recalcitrant SOC, a mechanism referred to as the priming effect (PE). Here, we performed series of short-term (7 days) incubation experiments to investigate whether the addition of (13C-labelled) algal carbon (C) affected SOC mineralization in an aquatic environment,

☐ DOI: https://link.springer.com/article/10.1007/s10533-026-01317-z

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250. Soil heterotrophic respiration after irrigation retirement is differentially influenced by moisture and substrate availability over time

☐ Biogeochemistry | 2026-03-09 [Latest] | 重要性: 14 分

☐ Abstract: Water limitations are forcing producers to transition large areas of currently irrigated farmland into dryland agriculture across the Western U.S. with unclear effects on global soil carbon (C) dynamics. An experiment established in 2017 in a no-till, maize system in Colorado suggested that soil heterotrophic respiration (Rh) following irrigation retirement was co-regulated by water and available C. We continued Rh measurements in 2021−2022 along with monthly soil samplings to explore the interactive effects of soil moisture and available C on microbial community composition and activity. Plant C inputs, available soil water, bacteria, fungi, and protozoa fatty acid methyl ester (FAME) biomarkers, enzyme activity, and Rh decreased after irrigation retirement, while actinobacteria abundance

☐ DOI: https://link.springer.com/article/10.1007/s10533-026-01312-4

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251. Precipitation legacy effects on gross N mineralization and nitrification rates in a Pinyon-Juniper dryland under altered precipitation

☐ Biogeochemistry | 2026-03-09 [Latest] | 重要性: 14 分

☐ Abstract: Changes in precipitation patterns can influence soil nitrogen (N) cycling by altering the timing and extent of moisture availability. Arid conditions, associated with more frequent and severe droughts, can increase soil inorganic N concentrations when diffusion constraints decouple N sources and sinks, making this N susceptible to loss. Further, shifts in precipitation patterns in one season can "carry over"N into subsequent seasons, known as precipitation legacies. However, how these legacies increase soil N availability through changes in inorganic N production (i.e., gross N mineralization and nitrification) remain unclear. We manipulated the amount and timing of precipitation in a pinyon–juniper dryland by either adding or excluding precipitation during the winter or summer months and

☐ DOI: https://link.springer.com/article/10.1007/s10533-026-01313-3

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252. Cover crops shape soil microbiome and ecological interactions in the soybean rhizosphere under long-term no-till

☐ Soil & Tillage Research [Latest] | 重要性: 11 分

☐ Abstract: Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Caio César Gomes Freitas, Carolinne Rosa de Carvalho, Nariane de Andrade, Rafael Lima Oliveira, Victor Lucas Vieira Prudêncio Araújo, Felipe Martins do Rêgo Barros, João Henrique dos Santos Ferreira, José Roberto Portugal, Carlos Alexandre Costa Crusciol, Fernando Dini Andreote

☐ DOI: https://www.sciencedirect.com/science/article/pii/S016719872600108X

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253. Mechanisms and sources of organic carbon accrual in deep soil under long-term straw return

☐ Soil & Tillage Research [Latest] | 重要性: 11 分

☐ Abstract: Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Wenhao Feng, Jie Zhou, Antonio Rafael Sánchez-Rodríguez, Yakov Kuzyakov, Ji Chen, Shang Wang, Haoruo Li, Jing Wang, Hongyuan Zhang, Roland Bol, Yuyi Li, Leanne Peixoto

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167198726001066

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254. Long-term N fertilization altered viral community composition and increased functional potentials of phosphorus cycling by causing soil acidification

☐ Soil Biology and Biochemistry [Latest] | 重要性: 9 分

☐ Abstract: Publication date: June 2026Source: Soil Biology and Biochemistry, Volume 217Author(s): Shuxun Cheng, Zhongmin Dai, Dan He, Ran Xue, Kankan Zhao, Jorge L. Mazza Rodrigues, Lucas W. Mendes, Scott X. Chang, Jianming Xu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0038071726000568

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255. Rapid assembly and functional differentiation of the soil surface microbiome in temperate agricultural soil

☐ Soil Biology and Biochemistry [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Soil Biology and Biochemistry, Volume 217Author(s): Christopher James O'Grady, Sally Hilton, Emma Picot, Sebastien Raguideau, Christopher Quince, Christopher J. van der Gast, Hendrik Schäfer, Gary D. Bending

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0038071726000489

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256. Building soil organic matter differing in physical fractions for fertilizer nitrogen retention

☐ Soil Biology and Biochemistry [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Soil Biology and Biochemistry, Volume 217Author(s): Quan Tang, Qingde Li, Xiaozhi Wang, Yuanyuan Huang, Tim J. Daniell, Enxuan Wu, Jing Wang, Feiyi Zhang, Zhenwang Li, Scott X. Chang, Zucong Cai, Yves Uwiragiye, Nyumah Fallah, Minggang Xu, Christoph Müller, Yi Cheng

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0038071726000556

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257. The Increasing Impact of Seasonality Biases on Model-Based Estimates of the Ocean Carbon Sink

☐ Global Biogeochemical Cycles | Sun, 15 Mar 2026 21: [Latest] | 重要性: 6 分

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008713

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258. Soil Moisture Reduction Accelerates CO2 Emissions in Near-Saturated Cold and Wet Ecosystems

☐ Global Biogeochemical Cycles | Fri, 13 Mar 2026 03: [Latest] | 重要性: 6 分

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008877

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259. Glacial Meltwater Promotes Biological Productivity and Subsequent Dissolved Organic Carbon Accumulation in the Eastern Ross Sea: Evidence From the Austral Summer of 2023

☐ Global Biogeochemical Cycles | Fri, 06 Mar 2026 01: [Latest] | 重要性: 6 分

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008770

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260. Harmonizing data from different soil organic matter fractionation protocols

☐ Geoderma [Latest] | 重要性: 6 分

☐ Abstract: Publication date: April 2026Source: Geoderma, Volume 468Author(s): Tchodjowiè P.I. Kpemoua, Marija Stojanova, Claire Chenu, Denis Angers, Amicie Delahaie, Jussi Heinonsalo, Kristiina Karhu, Marine Landrieux, Sam McNally, Cédric Plessis, Christopher Poeplau, Valérie Pouteau, Pierre Roudier, Eva Rabot, Juliette Roussia, Marcus Schiedung, Iñigo Virto, Pierre Barré

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0016706126001023

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261. Biochar-based fertilizer mitigates paddy soil N2O emissions by shifting nitrifier-denitrifier functional gene balance and enhancing recovery efficiency of nitrogen

☐ Soil & Tillage Research [Latest] | 重要性: 6 分

☐ Abstract: Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Delei Kong, Xianduo Zhang, Baogang Li, Gaodi Zhu, Mahaojiang Liu, Shuwei Liu, Yinxiu Liu, Peikun Jiang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167198726001212

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262. Synergistic optimization of plastic film mulching and nitrogen management and water-soil-greenhouse gas coupling mechanism on the Loess Plateau

☐ Soil & Tillage Research [Latest] | 重要性: 6 分

☐ Abstract: Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Peng Pan, Jin Shi, Congwei Sun, Modsarajah Rajendran, Enke Liu, Xurong Mei

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0167198726001078

【遥感与模型】

263. Quantifying Potential Nutrient Liberation Following Plant Investment in Root Exudates: Variation Across Soil Profiles and Ecosystems in the United States

☐ Journal of Geophysical Research - Biogeosciences | Fri, 13 Mar 2026 04: [Latest] | 重要性: 38 分

☐ Abstract: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009380

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264. Capturing Spatial Gradients of Water Color and Clarity in Subtropical Reservoirs During Drought

☐ Journal of Geophysical Research - Biogeosciences | Wed, 11 Mar 2026 04: [Latest] | 重要性: 34 分

☐ Abstract: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009401

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265. Contrasting CO2 Fluxes Among Arctic Pond Types Are Driven by Differences in Carbon Lability, Terrestrial Inputs, and Benthic Respiration

☐ Journal of Geophysical Research - Biogeosciences | Thu, 12 Mar 2026 06: [Latest] | 重要性: 32 分

☐ Abstract: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009318

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266. Gold-Standard Science in Uncertain Times —Thank You to the 2025 Reviewers of JGR Biogeosciences

☐ Journal of Geophysical Research - Biogeosciences | Sat, 14 Mar 2026 02: [Latest] | 重要性: 30 分

☐ Abstract: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009838

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267. Influence of Biofilms and Extracellular Polymeric Substances on High-Frequency Acoustic Backscatter

☐ Journal of Geophysical Research - Biogeosciences | Thu, 12 Mar 2026 06: [Latest] | 重要性: 30 分

☐ Abstract: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009446

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268. A mechanism-learning deeply coupled model for single-channel land surface temperature retrieval

☐ Remote Sensing of Environment [Latest] | 重要性: 13 分

☐ Abstract: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Tian Xie, Huanfeng Shen, Menghui Jiang, Huifang Li, Chao Zeng, Juan Carlos Jiménez-Muñoz, José A. Sobrino, Jun Ma, Guanhao Zhang, Liangpei Zhang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0034425726001203

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269. Pulse fragmentation-induced uncertainty in forest LAI mapping using UAV LiDAR

☐ Remote Sensing of Environment [Latest] | 重要性: 11 分

☐ Abstract: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Yuchen Bai, Philippe Verley, Tiangang Yin, Nicolas Lauret, Florence Forbes, Jean-Baptiste Durand, Grégoire Vincent

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0034425726001112

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270. Urban PolSAR decomposition: Challenges and new directions for emerging SAR technologies

☐ Remote Sensing of Environment [Latest] | 重要性: 9 分

☐ Abstract: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Dingfeng Duan, Yong Wang

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0034425726001331

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271. Spatial assessment of snow grain size from airborne lidar reflectance against coincident imaging spectroscopy retrievals

☐ Remote Sensing of Environment [Latest] | 重要性: 9 分

☐ Abstract: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Chelsea Ackroyd, Christopher P. Donahue, Brian Menounos, S. McKenzie Skiles

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0034425726001367

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272. Snow effects on altimeter waveforms over sea ice in the Weddell Sea - Part II: sea ice and snow spaceborne retrieval

☐ Remote Sensing of Environment [Latest] | 重要性: 9 分

☐ Abstract: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Lu Zhou, Henriette Skourup, Julienne Stroeve, Sahra Kacimi, Stefanie Arndt, Weixin Zhu, Alek Petty, Lanqing Huang, Shiming Xu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0034425726001306

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273. Phenology-Aligned multi-task temporal fusion framework for satellite-based triple-seasonal rice yield estimation in Southeast Asia

☐ International Journal of Applied Earth Observation [Latest] | 重要性: 9 分

☐ Abstract: Publication date: May 2026Source: International Journal of Applied Earth Observation and Geoinformation, Volume 149Author(s): Zhixian Lin, Kaiyu Guan, Sheng Wang, Qu Zhou, Liangzhi You, Xuan Chen, Xiangzhong Luo, Kejie Zhao

☐ DOI: https://www.sciencedirect.com/science/article/pii/S1569843226001470

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274. Modeling potential fish species richness within the Upper Paraguay River Basin in Brazil

☐ Ecological Modelling [Latest] | 重要性: 8 分

☐ Abstract: Publication date: June 2026Source: Ecological Modelling, Volume 516Author(s): Henrique Ledo Lopes Pinho, Marcelo Leandro Bueno, Karoline Aparecida de Sena, Julio César Jut Solórzano, Yzel Rondon Súarez

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0304380026000888

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275. Management approaches in decision support systems to mitigate the risk of natural disturbances in European temperate and boreal forests –a review

☐ Ecological Modelling [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Ecological Modelling, Volume 516Author(s): Adriano Mazziotta, Kyle Eyvindson, Katharina Albrich, Juha Honkaniemi, Joyce Machado Nunes Romeiro, Susanne Suvanto, Annika Kangas

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0304380026000943

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276. Operationalizing ecosystem service flows in land-use planning: Mitigating ecological risks through dynamic compensation

☐ Ecological Modelling [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Ecological Modelling, Volume 516Author(s): Yan Zhang, Lina Tang, Alimujiang Kasimu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0304380026001006

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277. Integrated system dynamics and life cycle assessment modeling for carbon mitigation pathways in China's industrial mariculture

☐ Ecological Modelling [Latest] | 重要性: 6 分

☐ Abstract: Publication date: June 2026Source: Ecological Modelling, Volume 516Author(s): Fengfan Han, Yang Bai, Zilun Yi, Fei Jia, Haochen Hou, Ying Liu

☐ DOI: https://www.sciencedirect.com/science/article/pii/S0304380026000980

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278. Real-time generation of gap-free MODIS leaf area index product from 2000 to 2024 using a deep learning method

☐ International Journal of Applied Earth Observation [Latest] | 重要性: 5 分

☐ Abstract: Publication date: April 2026Source: International Journal of Applied Earth Observation and Geoinformation, Volume 148Author(s): Guodong Zhang, Yimin Ni, Gang Sun, Gaofei Yin, Wei Zhao, Anxin Ding, Xinyan Liu, Yi Zhang, Jiangchuan Hu, Zongyan Li, Rui Chen, Meilian Wang, Aleixandre Verger

☐ DOI: https://www.sciencedirect.com/science/article/pii/S1569843226001561

【预印本】

279. When South meets North: a joint contact zone coinciding with environmental gradients in three boreal tree species

☐ bioRxiv Evolutionary Biology | 2026-03-17 [Latest] | 重要性: 40 分

☐ Abstract: Post-glacial recolonization of Fennoscandia created secondary contact zones in many species, offering opportunities to study how gene flow and selection contribute to their establishment and maintenance. Here, we analyse genomic data from three boreal tree species--Picea abies, Betula pendula, and Pinus sylvestris-sampled along a latitudinal gradient in Sweden. Despite differences in colonization timing and dispersal ecology, all three species exhibit north-south genetic structuring aligned with environmental gradients. Most notably, the two main genetic clusters within each species overlap in a shared contact zone, corresponding to the climatic transition between Sweden's two major environmental zones. The extent and structure of the contact zone differ among species: P. abies shows stron

☐ DOI: https://doi.org/10.1101/2026.03.13.711305

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280. A biosecurity baseline for transboundary management of marine biological invasions in the ROPME Sea Area

☐ bioRxiv Ecology | 2026-03-17 [Latest] | 重要性: 35 分

☐ Abstract: Marine and brackish-water ecosystems are increasingly degraded by cumulative human pressures, with biological invasions representing a major driver of biodiversity loss, ecosystem disruption, and socio-economic impacts. Effective management requires regionally harmonized and scientifically robust baselines capable of supporting coordinated transboundary decision-making. Here we present the first consolidated marine biosecurity baseline for the Regional Organization for the Protection of the Marine Environment (ROPME) Sea Area, a transboundary region characterized by extreme environmental conditions and increasing biosecurity pressure. A total of 192 species (123 extant and 69 horizon), including birds, fishes, tunicates, invertebrates, plants, and chromists, were systematically reviewed, t

☐ DOI: https://doi.org/10.1101/2026.03.13.711635

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281. Ancient transposable elements sustain global ecological adaptation despite chronically low nucleotide diversity

☐ bioRxiv Evolutionary Biology | 2026-03-17 [Latest] | 重要性: 25 分

☐ Abstract: Transposable elements (TEs) are dynamic components of eukaryotic genomes and a major source of structural and regulatory variation, yet their contribution to ecological adaptation over evolutionary time remains unresolved. This uncertainty is particularly acute in species that occupy broad environmental niches despite chronically low nucleotide diversity, where conventional models predict limited adaptive potential. Here we show that ancient TE polymorphisms underpin global ecological adaptation in Spirodela polyrhiza, one of the smallest flowering plants with low genome-wide nucleotide diversity. Most TE polymorphisms predate continental population divergence. Cold-season temperature emerges as the dominant selective axis, with adaptive signals overwhelmingly associated with TE polymorphi

☐ DOI: https://doi.org/10.1101/2026.03.14.711760

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282. Influence of ocean warming and acidification on juveniles of the true giant clam, Tridacna gigas, and its microalgal symbionts

☐ bioRxiv Zoology | 2026-03-13 [Latest] | 重要性: 22 分

☐ Abstract: Uncontrolled carbon dioxide emissions from human activities contribute to ocean warming and acidification. These alterations in ocean chemistry threaten marine organisms, such as the true giant clam, Tridacna gigas, which is already imperiled due to overharvesting and habitat destruction. To gain an understanding of the

physiological and molecular responses of T. gigas and its symbiotic dinoflagellates to ocean warming and acidification, we subjected juvenile individuals to different treatments simulating predicted seawater pH (7.6 and 8.0) and temperature (28{degrees}C, 30{degrees}C, 32{degrees}C and 34{degrees}C) levels for the next century. Juvenile giant clams were able to tolerate sustained exposure to temperatures of up to 32{degrees}C and pH as low as 7.6, while exposure to higher t

☐ DOI: https://doi.org/10.1101/2026.03.11.711206

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283. Disturbance and landscape characteristics interactively drive dispersal strategies in continuous and fragmented metacommunities

☐ bioRxiv Ecology | 2026-03-17 [Latest] | 重要性: 20 分

☐ Abstract: Habitat fragmentation, driven by human activities, disrupts habitat connectivity and alters ecological processes through geometric and demographic fragmentation effects. Dispersal plays a fundamental role in shaping the distribution, abundance, and persistence of species in modified landscapes. While previous research looked at the evolution of dispersal strategies at the species level, community-level dynamics remain underexplored. Species exhibit diverse dispersal strategies to persist in modified landscapes, yet predicting how these strategies interact at the community level requires a more integrated approach. This study employed an individual-based simulation model to explore how fragmentation and other landscape characteristics influence community-level dispersal strategies. We teste

☐ DOI: https://doi.org/10.1101/2026.03.13.711669

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284. Specialists drive biodiversity scaling in symbiotic relationships

☐ bioRxiv Ecology | 2026-03-17 [Latest] | 重要性: 18 分

☐ Abstract: The stunning diversity of symbiotic life forms, and their unique vulnerability to extinction, emerge from the close relationship between host and symbiont species richness. The general form of this relationship should be linear, but simulation studies have shown that it becomes sub-linear (and even power law-like) when sampling within a host-symbiont network. Here, we resolve this paradox with a new mathematical model of scaling in bipartite graphs, based on the independent behavior of specialist (single-host) and generalist (multi-host) symbionts. Using this model, we show that specialists constrain the architecture of ecological networks, and at global scales, drive the accumulation of symbiont biodiversity. By definition, specialists also face the highest risk of coextinction with their

☐ DOI: https://doi.org/10.1101/2026.03.13.711632

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285. Gene Expression Landscapes Driving Early Life Stages of the Keystone Seagrass Posidonia oceanica

☐ bioRxiv Plant Biology | 2026-03-13 [Latest] | 重要性: 18 分

☐ Abstract: Seagrasses are marine angiosperms forming extensive underwater meadows that provide habitat, stabilize sediments, store carbon, and protect coastlines. Posidonia oceanica is the endemic foundation seagrass species of the Mediterranean, yet its meadows are rapidly declining. Despite its ecological importance, the molecular basis of P. oceanica development remains poorly understood. Here, we analyzed gene expression in roots, leaves, and seeds across four developmental stages, revealing strong tissue-specific patterns and temporally regulated expression dynamics. Leaves exhibited active regulation of photosynthesis-related processes, while roots were enriched in pathways linked to carbohydrate metabolism and cell wall biogenesis, supporting primary root growth and establishment. Seeds retain

☐ DOI: https://doi.org/10.1101/2026.03.13.711526

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286. In vivo interrogation of transcriptional and epigenetic regulators of lung epithelial regeneration

☐ bioRxiv Genomics | 2026-03-17 [Latest] | 重要性: 13 分

☐ Abstract: Effective alveolar repair after viral lung injury requires precise coordination of alveolar type 2 cell (AT2) proliferation and differentiation to restore lung function. To uncover causal regulators of this process in the native tissue environment, we developed SAGE (Stable Adeno-Associated Virus Genomic IntEgration), an engineered AAV system that enables high-throughput in vivo genetic interrogation. SAGE supports both bulk phenotypic screening (SAGE-Perturb) and single-cell transcriptomic profiling (SAGE-Perturb-seq). Using this approach, we identified lysine acetyltransferase 8 (Kat8) as essential for epithelial repair following viral infection through the Non-Specific-Lethal (NSL) complex, and generated a time-resolved, high-resolution functional map of transcription factor knockouts d

☐ DOI: https://doi.org/10.1101/2026.03.13.711474

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287. Genome-wide Identification of Transcriptional Start Sites and Candidate Enhancers Regulating Worker Metamorphosis in Apis mellifera

☐ bioRxiv Genomics | 2026-03-16 [Latest] | 重要性: 13 分

☐ Abstract: Eusociality in bees represents a major evolutionary transition and understanding its molecular basis is fundamental for sociogenomic studies. Comparative genomics has revealed correlations between transcription factor binding site (TFBS) abundance and social complexity; however, when and where these TFBSs function in a eusocial context remains largely unclear. In this study, we performed cap analysis of gene expression (CAGE) during worker metamorphosis in the honeybee Apis mellifera to identify TFBSs within active enhancers and decipher the regulatory relationships between these enhancers and their target genes. We identified 17,349 transcription start sites (TSSs) and 842 candidate enhancers. Using CAGE, we identified five clusters based on expression patterns. Notably, genes associated

☐ DOI: https://doi.org/10.1101/2026.03.12.711487

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288. AI for Fisheries Science: Neural Network Tools for Forecasting, Spatial Standardization, and Policy Optimization

☐ bioRxiv Ecology | 2026-03-17 [Latest] | 重要性: 10 分

☐ Abstract: The development of Artificial Intelligence (AI) presents novel opportunities for tackling complex marine resource management challenges. Among AI models, neural networks are a powerful class of tools capable of learning nonlocal and lagged patterns from fisheries data as well as approximating nonlinear relationships among multiple latent variables using estimation methods that automatically implement statistical shrinkage. This gives them potential to effectively handle data obtained from fisheries populations subject to dynamic environments. We highlight two flexible subclasses and one application of neural networks: Long Short-Term Memory (LSTM) and Convolutional Neural networks (CNNs) and policy discovery via Reinforcement Learning. LSTMs are designed to handle sequential data by allowi

☐ DOI: https://doi.org/10.1101/2026.03.13.711664

289. Shifting forage selection subsidizes seasonal resource scarcity

☐ bioRxiv Ecology | 2026-03-17 [Latest] | 重要性: 10 分

☐ Abstract: Nitrogen (N) is limiting for terrestrial herbivores, particularly over winter. Caribou (Rangifer tarandus) have adapted to seasonal scarcity of N by accruing muscle mass during the growing season when N is more abundant. Nitrogen stored in muscle tissue is then relied upon during winter to compensate for dietary deficits. Once their diet shifts from N-rich vascular plants to N-poor lichen over winter, caribou can lose ~30% of their muscle mass. As catabolized N is shed in urine on wintering grounds, caribou could act as elemental transport across seasons and landscapes. Furthermore, if deposited N is taken up by lichen or other winter forage, it might enrich the nitrogen-poor winter diet of caribou in the future. We tested this potential transport via three steps. We analysed Cladonia spp.

☐ DOI: https://doi.org/10.1101/2026.03.13.711571

290. Single-library chromosome-scale diploid assemblies of vole genomes resolve a species-specific duplication implicated in pair bonding

☐ bioRxiv Genomics | 2026-03-17 [Latest] | 重要性: 9 分

☐ Abstract: High-quality reference genomes are essential to effectively characterize genomic drivers of speciation, phenotypic diversity, and disease causality. Larger complex genomes often require integration of long-read DNA sequencing with additional genomic data, such as chromosome conformation capture (Hi-C or CiFi) to generate phased chromosome-scale assemblies, however this requires multiple sequencing platforms (in the case of Hi-C) or the construction of multiple long-read sequencing libraries. Here, we devise a strategy that combines PacBio HiFi and CiFi sequencing in a single library and run to efficiently produce high-quality contiguous chromosome-scale diploid genome assemblies. We apply this approach to liver tissue from single individuals of prairie vole (Microtus ochrogaster) and meado

☐ DOI: https://doi.org/10.1101/2026.03.13.711624

291. Chromatin dynamics identifies 78 genes at loci associated with elevated intraocular pressure and primary open-angle glaucoma

☐ bioRxiv Genomics | 2026-03-16 [Latest] | 重要性: 9 分

☐ Abstract: Primary open-angle glaucoma (POAG) is a chronic neurodegenerative disorder, and elevated intraocular pressure (IOP) represents the major, and only modifiable, risk factor for the disease. We modeled increased IOP by treating three primary human trabecular meshwork (TM) cell strains with dexamethasone, then generated a high-resolution map of promoter-centered chromatin contacts and regulatory modules to decipher how the genomic architecture and epigenetic state of disease-associated loci contribute to pathogenesis. We identify dynamic changes in chromatin compartments and looping, cis-regulatory elements and transcription factor hubs corresponding to altered transcriptional profile. By integrating GWAS-associated variants with dexamethasone-induced 3D chromatin landscape, we discovered 26 l

☐ DOI: https://doi.org/10.1101/2026.03.12.711121

292. Horizontal transfer promotes allele segregation in multicopy plasmids

☐ bioRxiv Evolutionary Biology | 2026-03-17 [Latest] | 重要性: 8 分

☐ Abstract: Plasmids reside within prokaryotic cells in multiple copies and can spread horizontally between hosts. Their multicopy nature enables intracellular allele diversity (heteroplasmy), and their segregation depends on the modes of plasmid replication and partition. Horizontal plasmid transfer has the potential to alter plasmid allele composition, however its impact on plasmid allele dynamics remains poorly understood. Here, we show that conjugative plasmid transfer accelerates the segregation of plasmid heteroplasmy by promoting the emergence of homoplasmic hosts. Using a quantitative experimental system to track plasmid allele dynamics in Acinetobacter baylyi under non-selective conditions, we followed the fate of a novel antibiotic resistance allele introduced into an ancestral donor populat

☐ DOI: https://doi.org/10.1101/2026.03.13.711586

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293. Estimating the evolutionary fitness of specific synonymous codon changes

☐ bioRxiv Evolutionary Biology | 2026-03-17 [Latest] | 重要性: 8 分

☐ Abstract: Synonymous mutations do not alter proteins but undergo natural selection in many species. For Drosophila melanogaster , reports on selection strength vary from undetectable to surprisingly strong. Here we apply a new method to estimate the population selection coefficient (2Ns) for all 134 ordered pairs of synonymous codon changes. The method uses ratios of site frequency spectra (SFS) for codon changes to neutral changes, and does not depend on divergence data, or codon frequencies, and is relatively insensitive to demographic history. Results indicate that natural selection on synonymous codons is weak, with $|2Ns| < 2.07$ for all pairs of codons and $|2Ns| < 1$ for 64% of codon changes. Despite being derived solely from polymorphism data, codon fitness estimates are strongly correlated with

☐ DOI: https://doi.org/10.1101/2026.03.16.712166

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294. Identification of the Phytophthora PAMP Pep-13 Receptor Using Diploid Potato Inbred Lines

☐ bioRxiv Plant Biology | 2026-03-16 [Latest] | 重要性: 8 分

☐ Abstract: Plants employ cell surface receptors to recognize pathogen-associated molecular patterns (PAMPs) and activate pattern-triggered immunity, a crucial defense mechanism against invading pathogens. Pep-13 is a PAMP derived from a class of conserved cell wall transglutaminases present in Phytophthora species, and its receptor PERU was reported recently. In our parallel study, we observed distinct responses to Pep-13 between two diploid potato inbred lines: E454 recognizes Pep-13, whereas A018 does not. Genetic analysis demonstrated that Pep-13 recognition in E454 is controlled by a single genetic locus, tentatively designated TGER (Transglutaminase elicitor response). Through bulked segregant analysis sequencing, followed by complementation assays, we found that the TGERa gene in E454 is essent

☐ DOI: https://doi.org/10.1101/2026.03.15.709221

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295. Using insertable cardiac monitors to test determinants of heart rate and activity in captive baboons

☐ bioRxiv Zoology | 2026-03-17 [Latest] | 重要性: 8 分

☐ Abstract: Background: Insertable cardiac monitors (ICMs) provide fine-grained, continuous data on cardiac activity. These data have great potential to reveal individual physiology, energetics, and stress responses, with implications for animal health, cognition, welfare, and conservation. However, these devices must be tested for safety, accuracy, and biological validity before being deployed in new species. Here we do so for the Reveal LINQTM ICM (Medtronic, Minneapolis, MN USA) over an 8-month period in 10 adult female baboons (Papio anubis and P. cynocephalus) at the Kenya Institute of Primate Research in Kenya. We also report data on heart rate, physical activity, and body temperature in unrestrained, conscious, captive baboons during their normal activities. Finally, we test how heart rate and

☐ DOI: https://doi.org/10.1101/2026.03.13.710869

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296. Reliable quantification of multiplexed genetically encoded biosensors responsiveness in plant tissues

☐ bioRxiv Plant Biology | 2026-03-16 [Latest] | 重要性: 6 分

☐ Abstract: Genetically encoded biosensors are one of essential tools in biological research. They enable visualization of molecules of interest from the subcellular level to entire organism level in vivo and can be used to monitor presence of small molecules, gene expression, protein activity, and protein degradation. However, multiplexing fluorescent biosensors in plants is notoriously difficult due to signal bleed-through and strong autofluorescence from chlorophyll. In this study, we investigated the potential of multiplexing biosensors based on the selection of reporter fluorescent proteins. We characterized the emission spectra, fluorescence lifetimes, and relative brightness of diverse fluorescent proteins in plant leaves. We show that selected proteins exhibit comparable brightness, supporting

☐ DOI: https://doi.org/10.1101/2026.03.13.711581

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297. In silico analysis reveals the structural basis of TomEP specificity, a tomato extensin peroxidase

☐ bioRxiv Plant Biology | 2026-03-13 [Latest] | 重要性: 6 分

☐ Abstract: BackgroundExtensin peroxidases (EPs) are class III plant peroxidases and are responsible for intermolecular covalent crosslinking of extensin (EXT) monomers to create scaffolds within plant cell walls. The formation of these scaffolds impacts plant development, mechanical wounding, and response to pathogen attacks. Therefore, elucidating the molecular mechanism controlling covalent crosslinking of EXT monomers is crucial for understanding cell wall deposition and potentially improving plant growth and adaptation. The focus of this work is to use in silico analysis to determine the structural characteristics of an EP from tomato (TomEP) to elucidate its specificity for crosslinking of EXT monomers. ResultsIn this study the two-dimensional (2D) and three-dimensional (3D) structures of TomEP

☐ DOI: https://doi.org/10.1101/2026.03.10.710923

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298. Potential acoustic signatures of stress in black soldier fly (Hermetia illucens; Diptera: Stratiomyidae) larvae

☐ bioRxiv Zoology | 2026-03-10 [Latest] | 重要性: 6 分

☐ Abstract: Black soldier fly larvae (BSFL) have quickly become one of the most farmed animals in the world. However, little is known about how to monitor stress and welfare in these animals. The difficulty of welfare assessment is compounded by the fact that BSFL live in their feed and prefer darkness. This behaviour makes it challenging to observe potential welfare indicators without inducing stress via disturbing the larvae or moving them into the light. However, acoustic devices may be able to pick up signatures of stress in the population even while they are out of sight, allowing for remote monitoring of animals in natural conditions (in the feed and/or in the dark). Acoustic monitoring of this type has been deployed for the detection of insects in stored grains, suggesting this method holds som

☐ DOI: https://doi.org/10.1101/2026.03.06.709542

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299. Resilience-Aware Energy Management of Microgrids Incorporating Dynamic Thermal Line Rating

☐ Research Square - Environment | 2026-03-17 19:31:45 [Latest] | 重要性: 6 分

☐ Abstract: The growing infiltration of renewable energy sources, these bidirectional flows, and disturbance-induced operation have posed a considerable challenge in the distribution-level microgrids, where the constraints of the typical energy management system (EMS) based on static network constraints have been exposed. Specifically, the thermal characteristics of distribution lines are so frequently overlooked. The paper suggests a resilience-conscious microgrid energy management system that directly includes dynamic thermal line rating (TLR) as one of the key operational constraints. A hybrid learning-optimization approach is designed, with the integration of Long Short-Term Memory (LSTM) based forecasting of renewable generation, load demand, and conductor temperature with a mixed-integer linear

☐ DOI: https://www.researchsquare.com/article/rs-8838433/latest

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300. In vivo multiomic Perturb-seq with enhanced nuclear gRNA capture

☐ bioRxiv Genomics | 2026-03-17 [Latest] | 重要性: 5 分

☐ Abstract: In vivo CRISPR screening with joint transcriptomic and chromatin readouts has been limited by inefficient recovery of gRNAs from nuclei. Here, we developed in vivo multiomic Perturb-seq, an effective platform combining nuclear transcript anchoring with gRNA-specific capture and amplification to enable high-fidelity, high-recovery gRNA assignment and scalable perturbation-resolved single-nucleus multiomics. Applying this platform to interrogate neurodevelopmental disorder risk genes in the developing cortex reveals cell-type-specific transcriptomic and epigenomic perturbation phenotypes.

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